

# CSE 6740: Computational Data Analysis

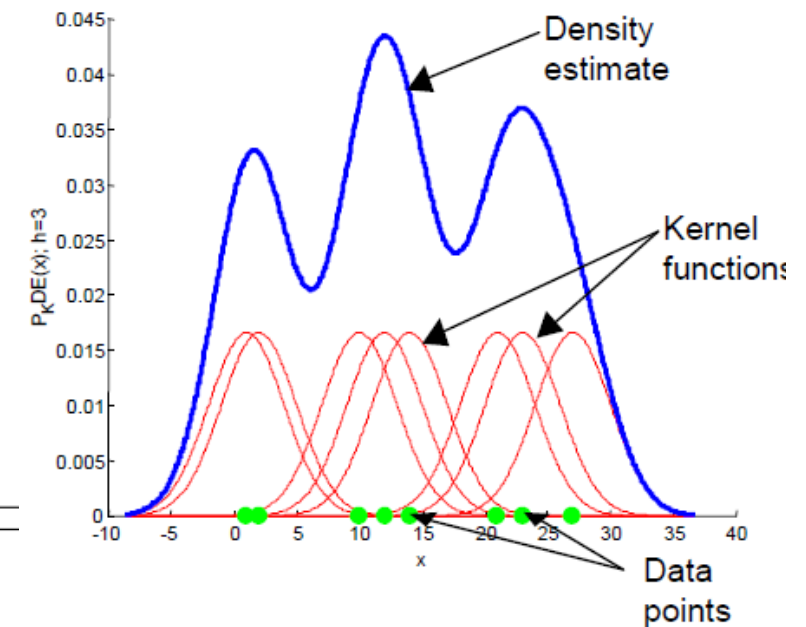
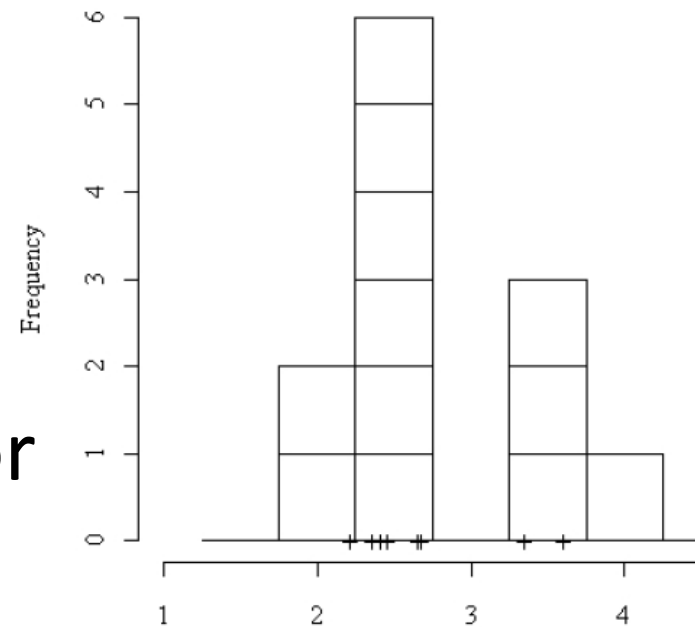
## Spring 2026

# Density Estimation and Gaussian Mixture Model

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02/10

# Nonparametric models

- Eg. Histogram
- Eg. Kernel density estimator



- What are nonparametric models?
  - “nonparametric” does **not** mean there are no parameters
  - can not be described by a fixed number of parameters
  - one can think of there are many many parameters

# 1-D Histogram

- One the simplest nonparametric density estimator

- Given  $m$  iid samples  $\mathcal{D} = \{x^1, x^2, \dots, x^m\}, x^i \in [0,1)$

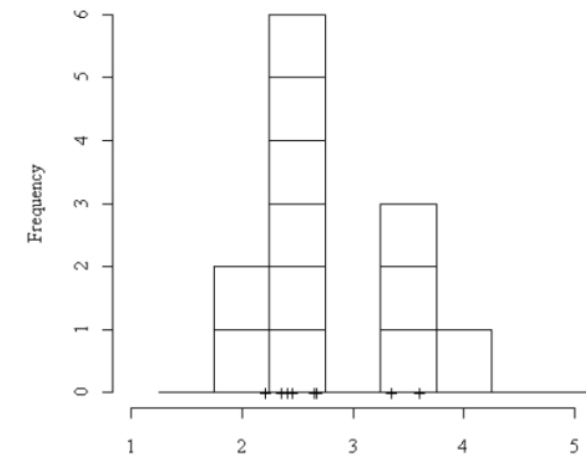
- Split  $[0,1)$  into  $n$  bins

$$B_1 = \left[0, \frac{1}{n}\right), B_2 = \left[\frac{1}{n}, \frac{2}{n}\right), \dots, B_n = \left[\frac{n-1}{n}, 1\right)$$

- Count the number of points,  $c_1$  within  $B_1$ ,  $c_2$  within  $B_2$ ...

- For a new test point  $x$

$$p(x) = \sum_{j=1}^n \frac{nc_j}{m} I(x \in B_j)$$



# Why is histogram valid?

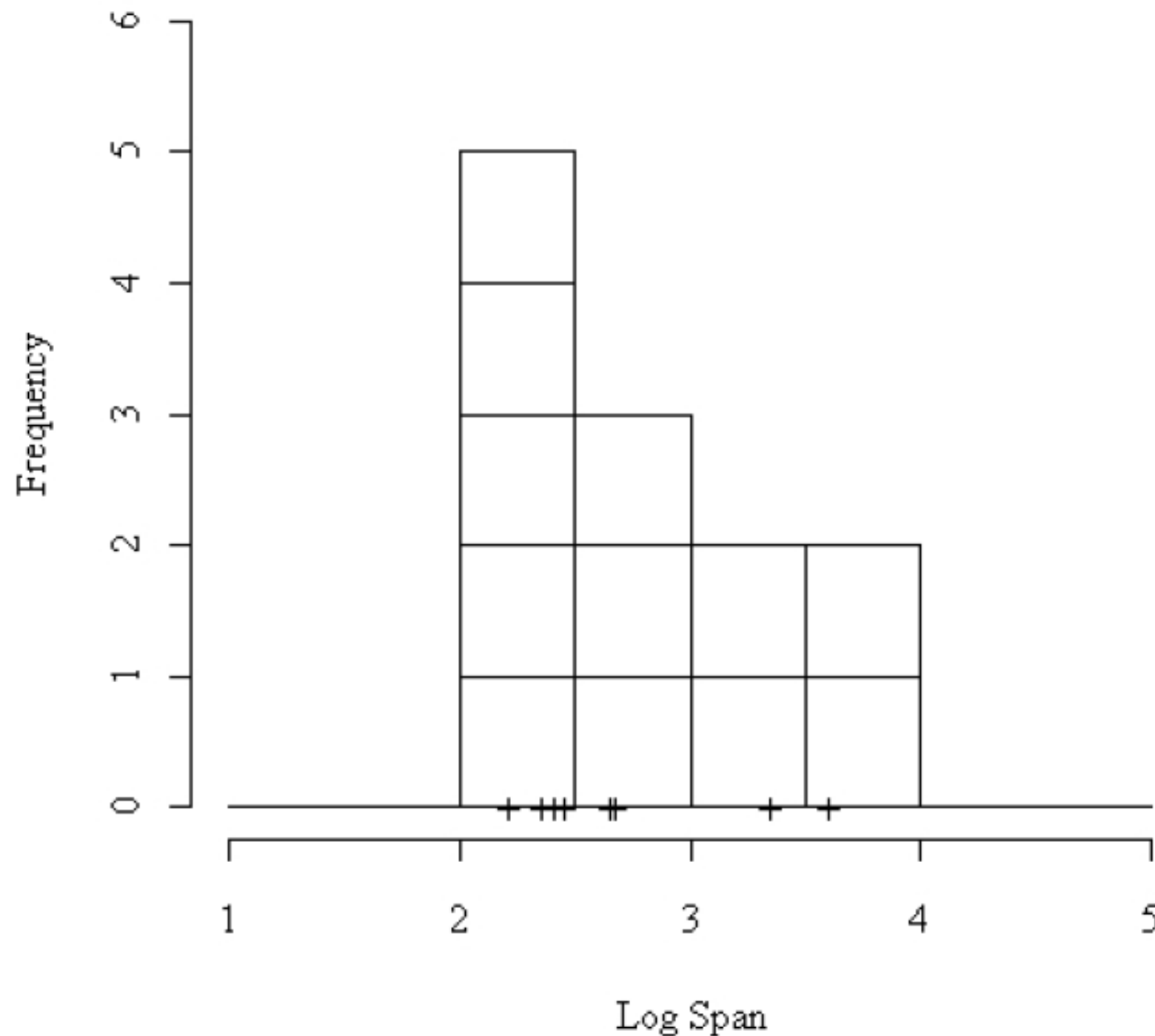
- Requirement for density  $p(x)$
- $p(x) \geq 0, \int_{\Omega} p(x)dx = 1$

- For histogram,

$$\begin{aligned}\int_{[0,1)} p(x)dx &= \int_{[0,1)} \sum_{j=1}^n \frac{nc_j}{m} I(x \in B_j) dx \\ &= \sum_{j=1}^n \int_{\left[\frac{j-1}{n}, \frac{j}{n}\right)} \frac{nc_j}{m} dx \\ &= \sum_{j=1}^n \frac{c_j}{m} = 1\end{aligned}$$

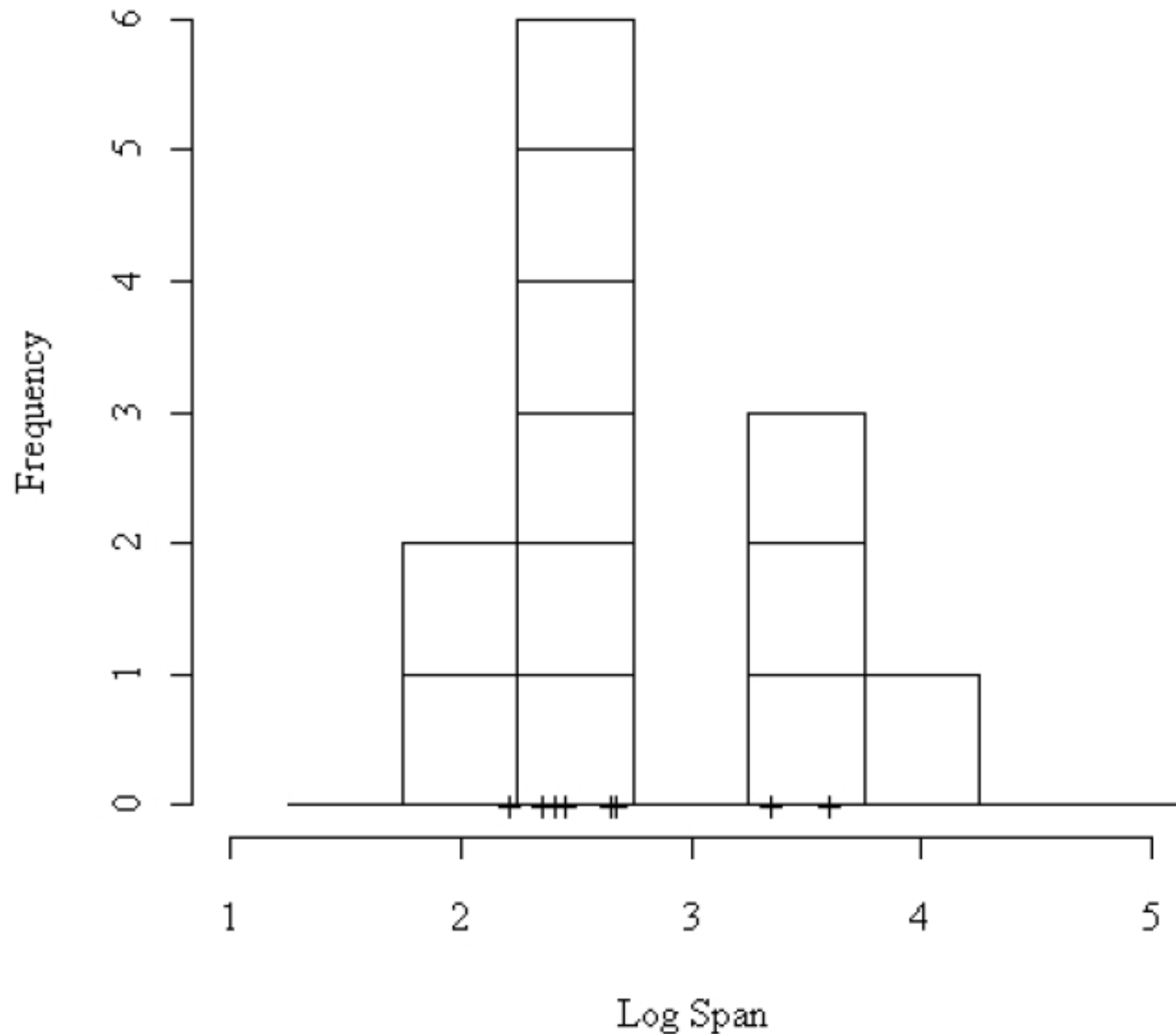
# Output depends on where you put the bins

**Histogram with breaks at n.0 and n.5  
binwidth=0.5**



# Output depends on where you put the bins

**Histogram with breaks at n.25 and n.75  
binwidth=0.5**



# Higher dimensional histogram

- Given  $m$  iid samples  $\mathcal{D} = \{x^1, x^2, \dots, x^m\}$ ,  $x^i \in [0,1)^d$

- Split  $[0,1)^d$  evenly into  $n^d$  bins

$$B_1 = \left[0, \frac{1}{n}\right) \times \left[0, \frac{1}{n}\right) \dots \times \left[0, \frac{1}{n}\right),$$

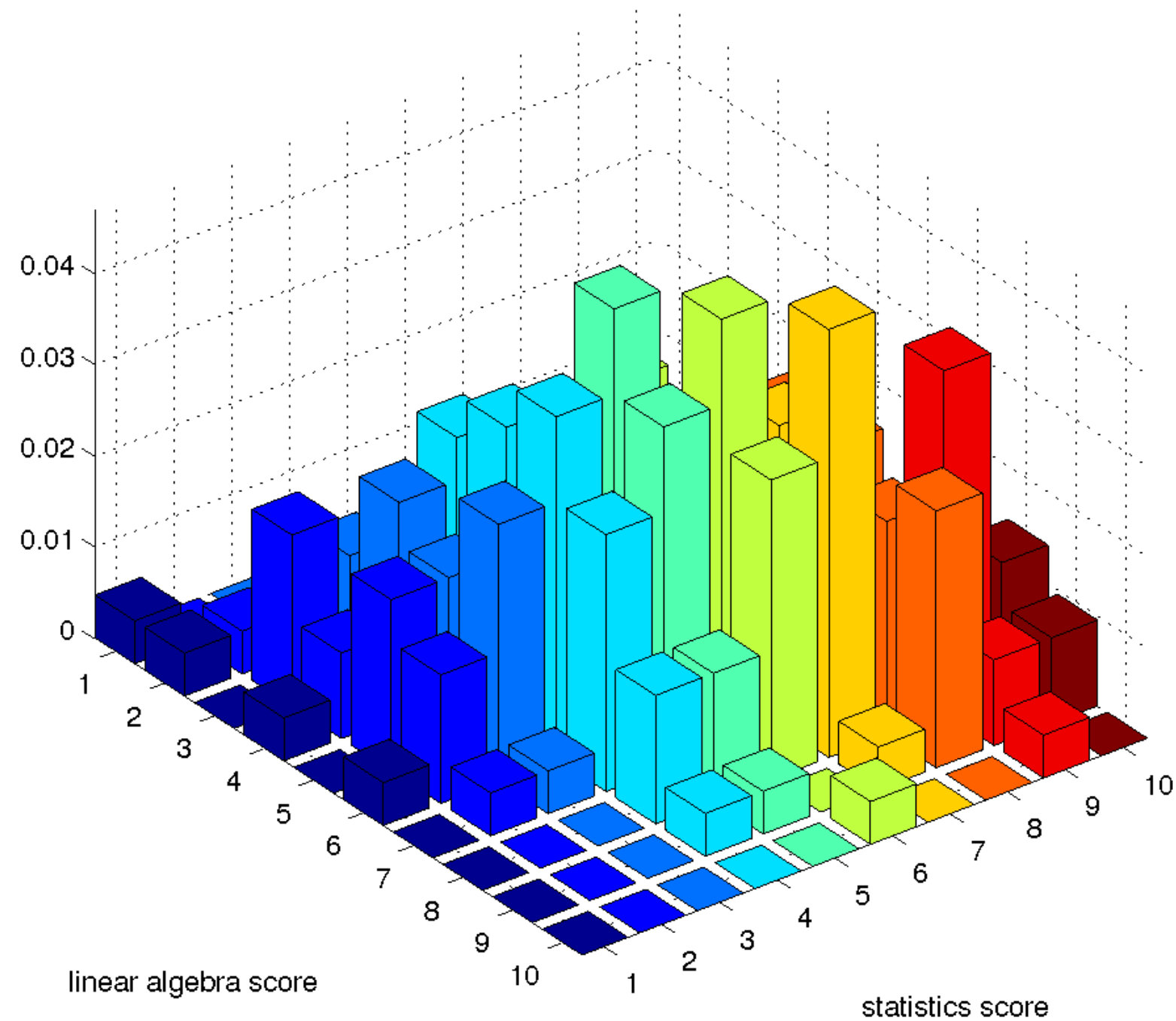
$$B_2 = \left[\frac{1}{n}, \frac{2}{n}\right) \times \left[0, \frac{1}{n}\right) \dots \times \left[0, \frac{1}{n}\right),$$

$$\dots$$
$$B_{n^d} = \left[\frac{n-1}{n}, 1\right) \times \left[\frac{n-1}{n}, 1\right) \dots \times \left[\frac{n-1}{n}, 1\right)$$

- Bin size is  $h = \frac{1}{n}$



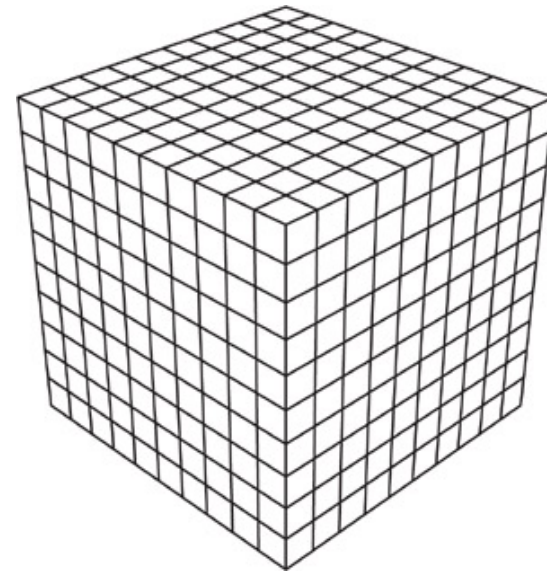
# Class scores





# Computation and statistical considerations

- Problem I: **too many bin! Not good for high dimensional data**
  - If  $n^d$  is larger than  $m$ , most bins are empty
  - Eg.  $n = 10, d = 6$ , need  $\sim 1$  million data points



- Problem II: statistically histogram is not the best

- Integrated risk:

$$r(\hat{p}, p) := \int_R \mathbb{E}_X \left[ (\hat{p}(x) - p(x))^2 \right] dx$$

- Histogram (with bin size  $h \sim m^{-1/3}$ )

$$r(\hat{p}, p) \sim \frac{C}{m^{2/3}}$$

- Kernel density estimator (with bandwidth  $h \sim m^{-1/5}$ )

$$r(\hat{p}, p) \sim \frac{C}{m^{4/5}}$$

- Difference even big for high dimensional data

# Kernel density estimation

- Kernel density estimator

$$p(x) = \frac{1}{m} \sum_i^m \frac{1}{h} K\left(\frac{x^i - x}{h}\right)$$

- Smoothing kernel function

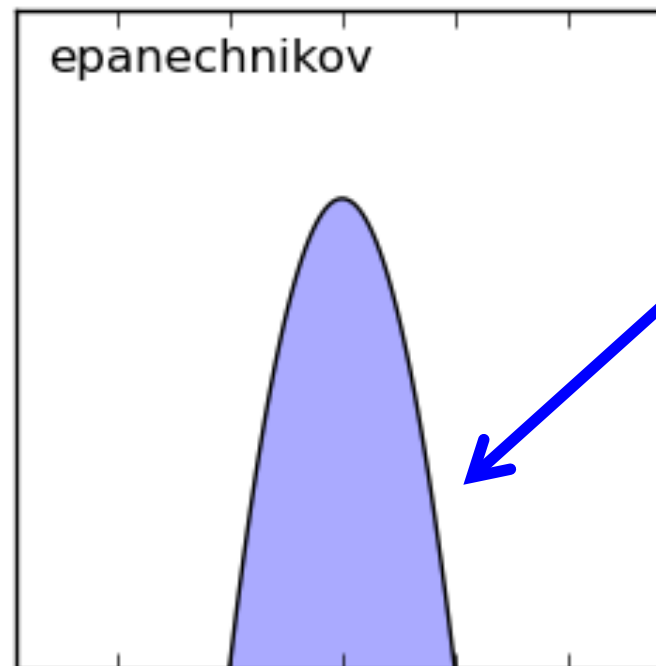
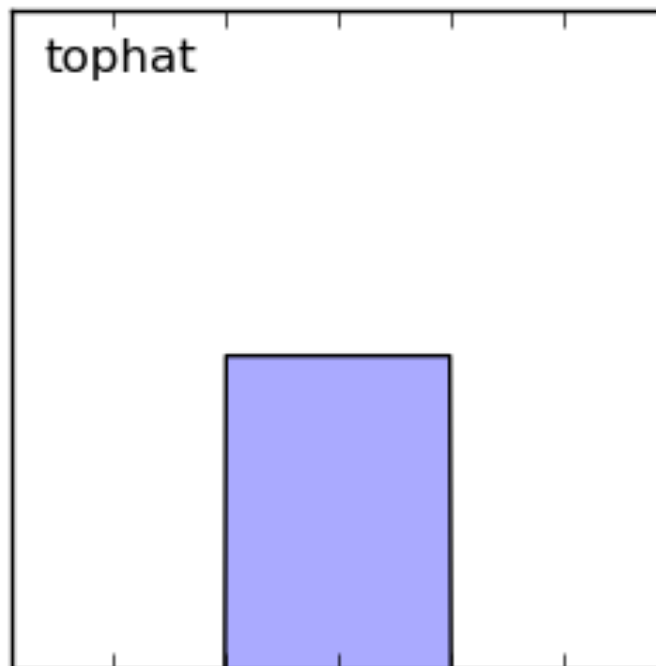
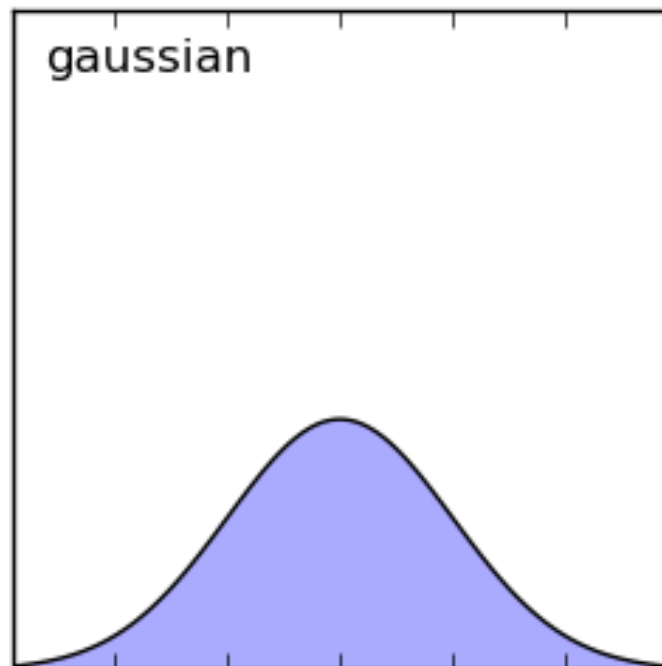
- $K(u) \geq 0$ ,
- $\int K(u)du = 1$ ,
- $\int uK(u) = 0$ ,
- $\int u^2 K(u)du < \infty$

- An example: Gaussian kernel  $K(u) = \frac{1}{\sqrt{2\pi}} e^{-u^2/2}$

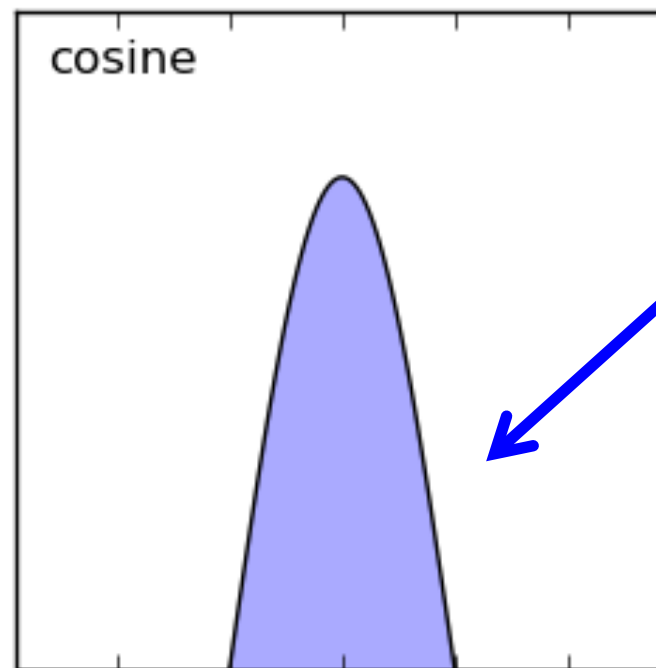
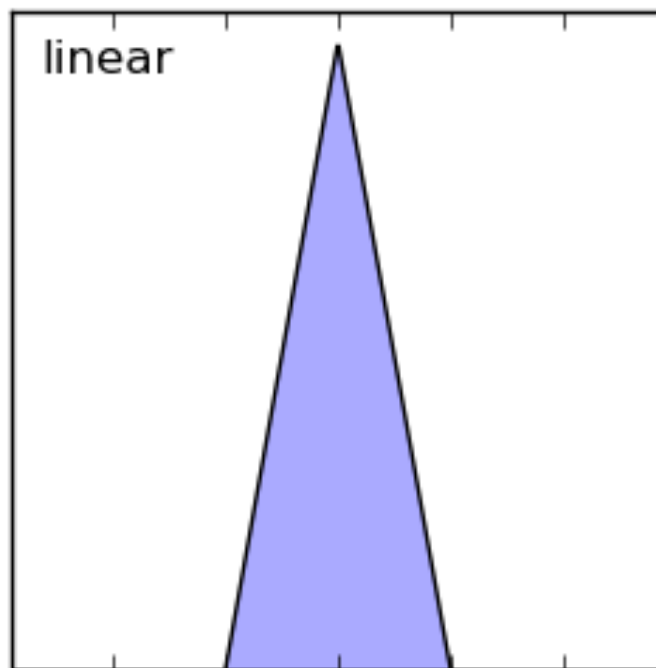
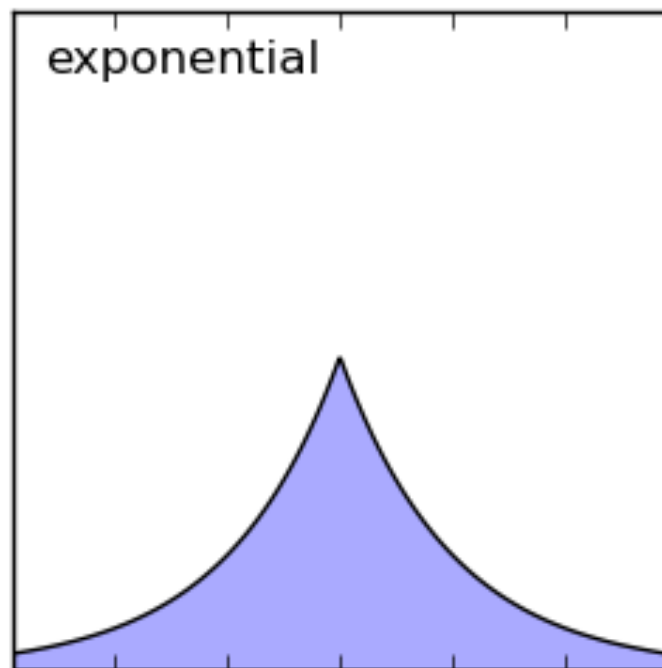
# Smoothing kernel functions

- An example: Gaussian kernel  $K(u) = \frac{1}{\sqrt{2\pi}} e^{-u^2/2}$

Available Kernels

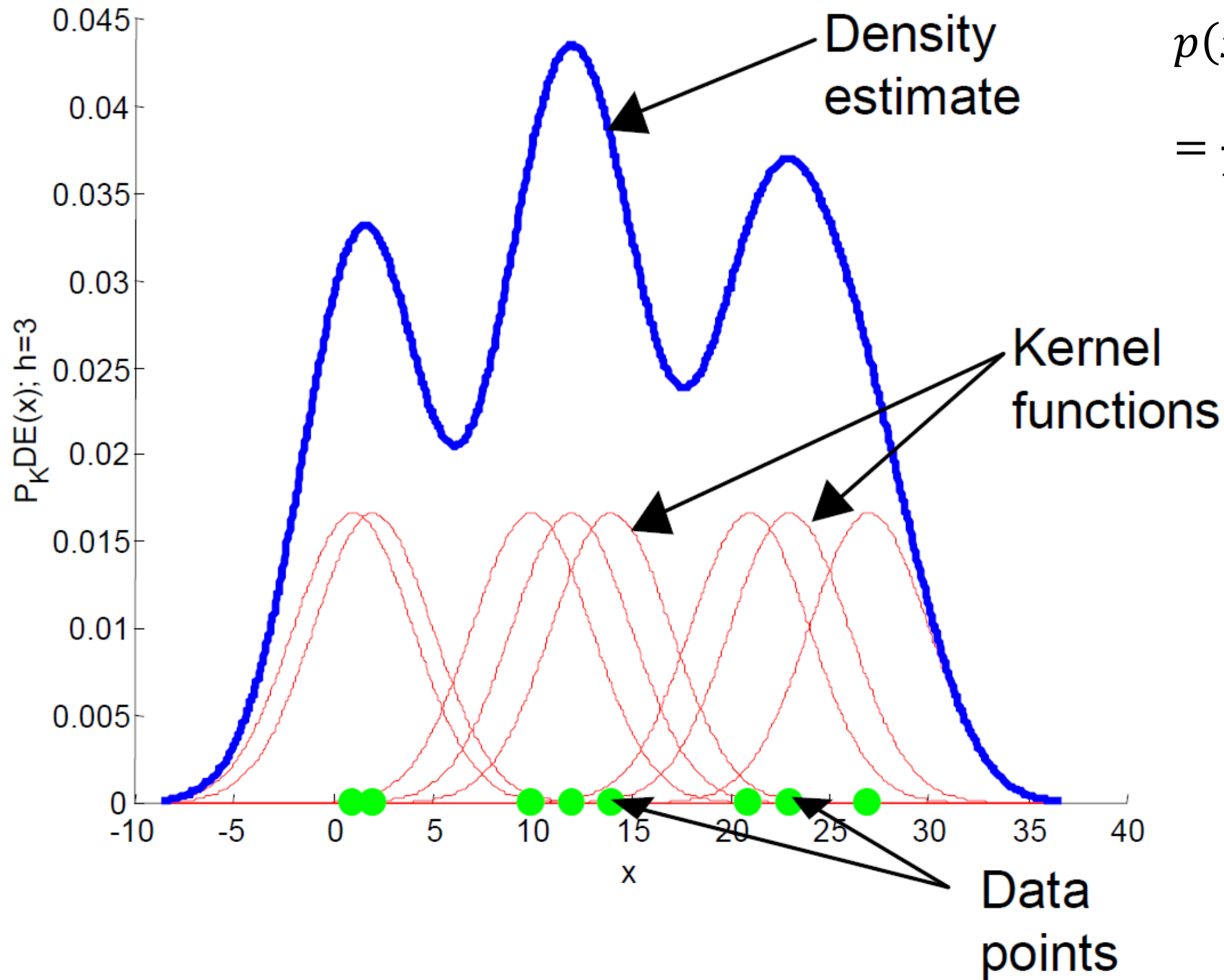


$$K(u) = \frac{3}{4}(1 - u^2)I(|u| \leq 1)$$



$$K(u) = \frac{\pi}{4} \cos\left(\frac{\pi}{2}u\right)I(|u| \leq 1)$$

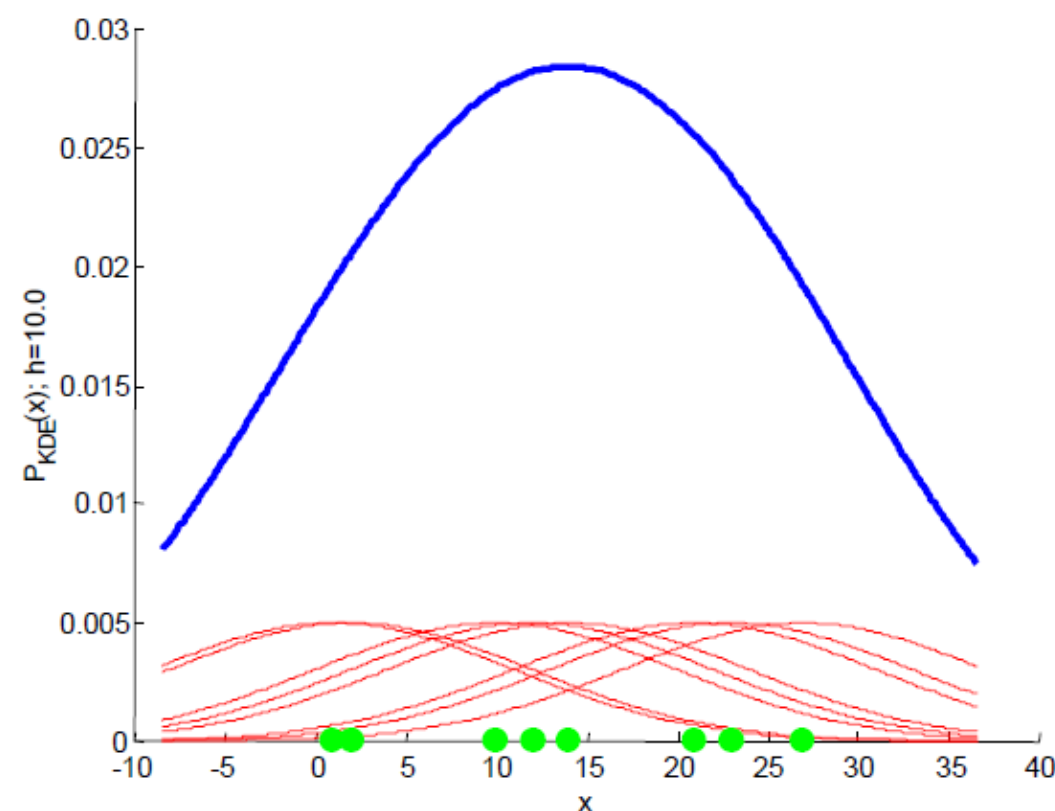
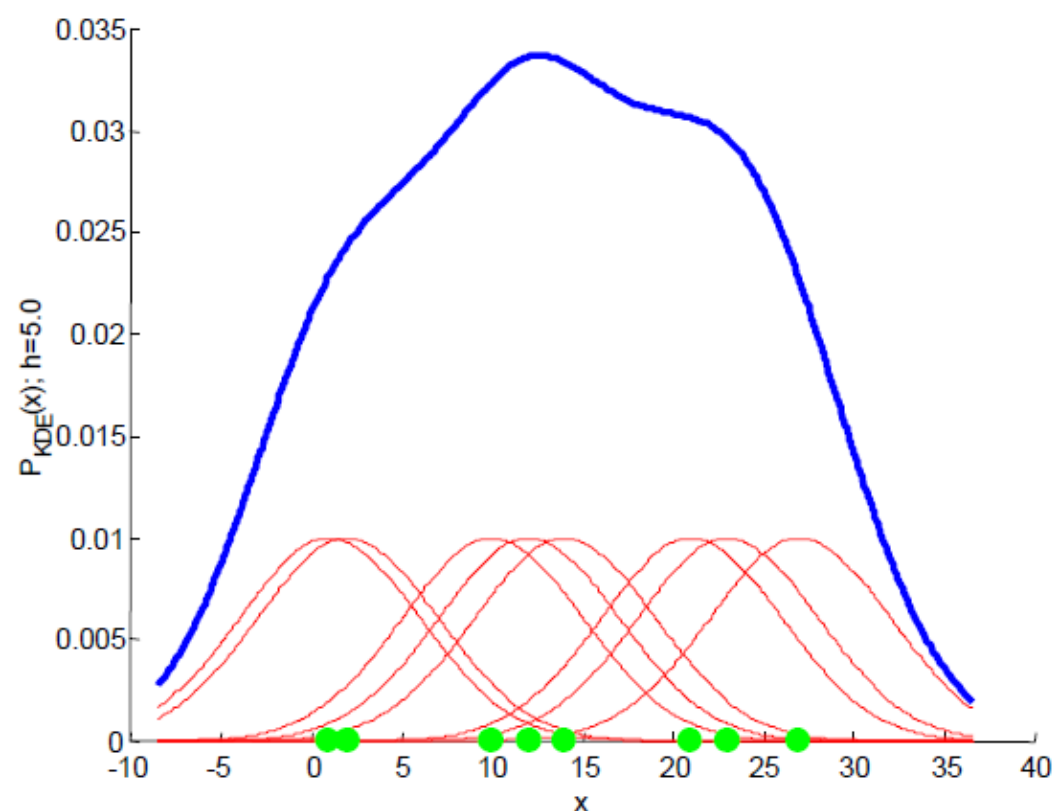
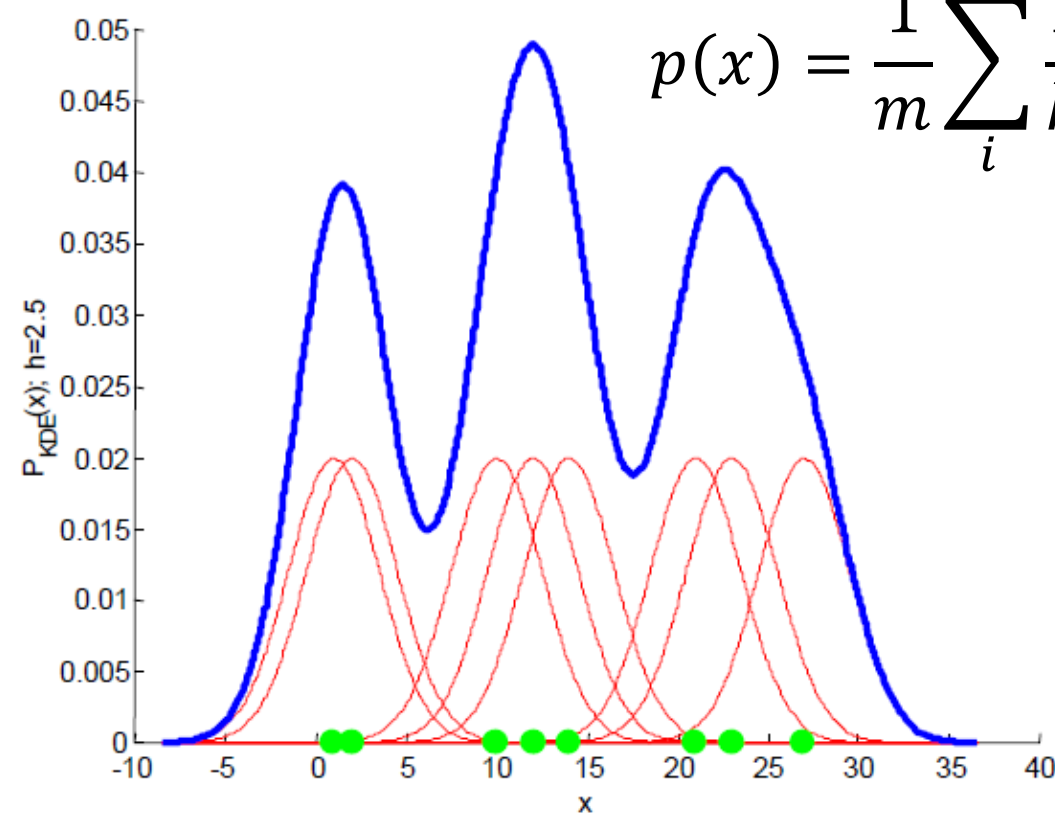
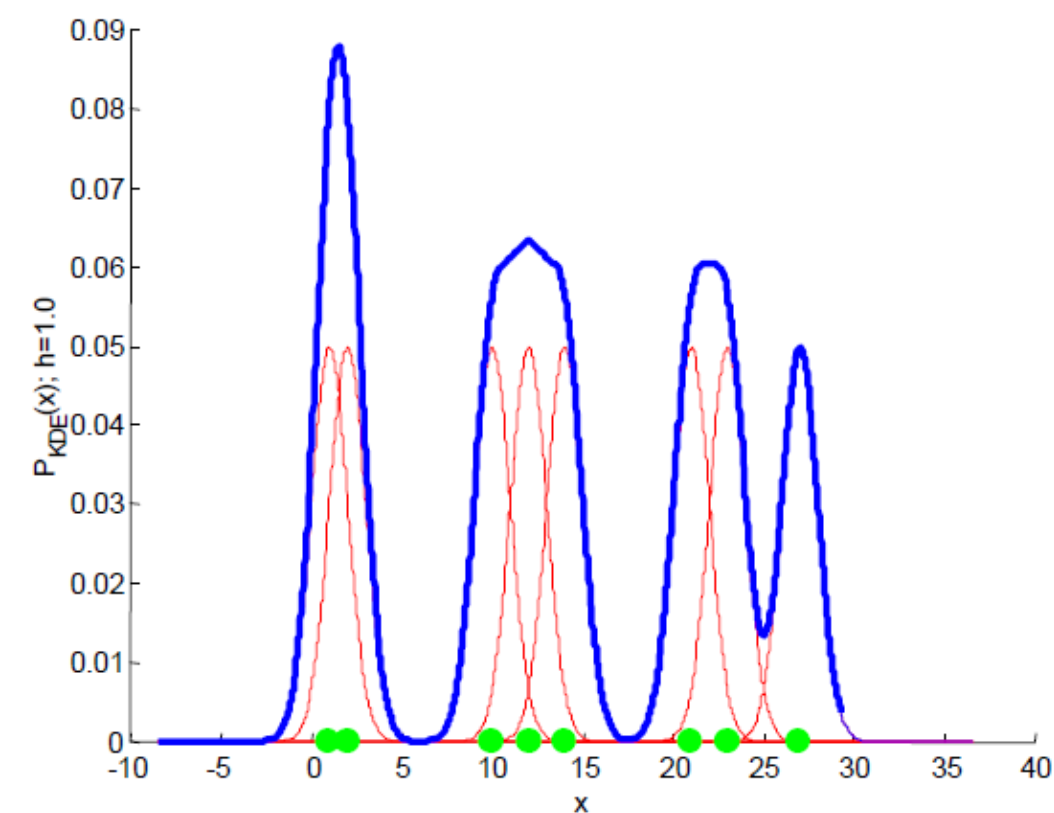
# Example



$$p(x) = \frac{1}{m} \sum_i^m \frac{1}{h} K\left(\frac{x^i - x}{h}\right)$$

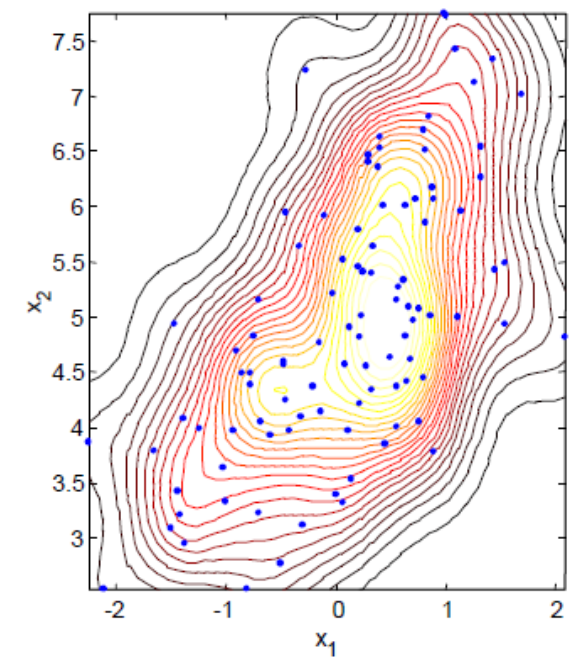
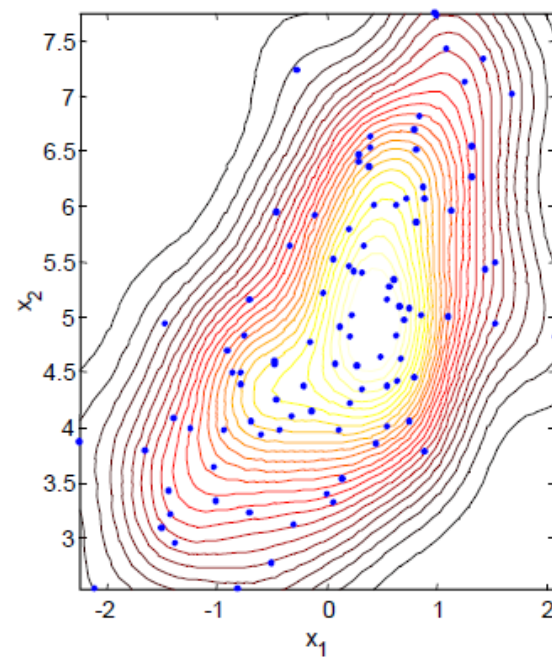
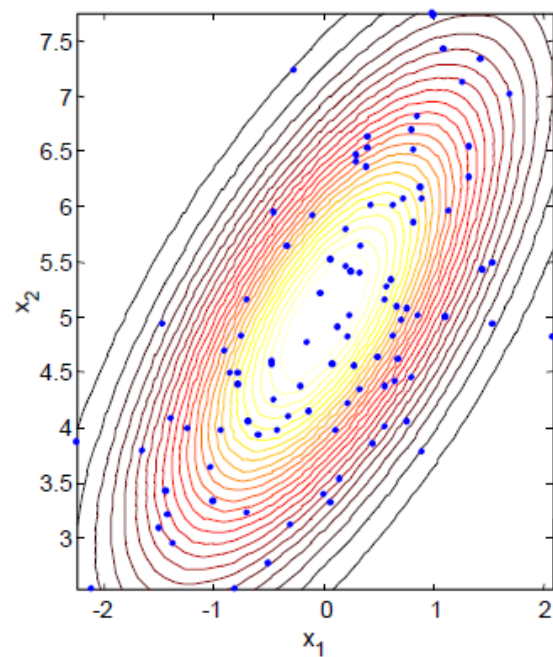
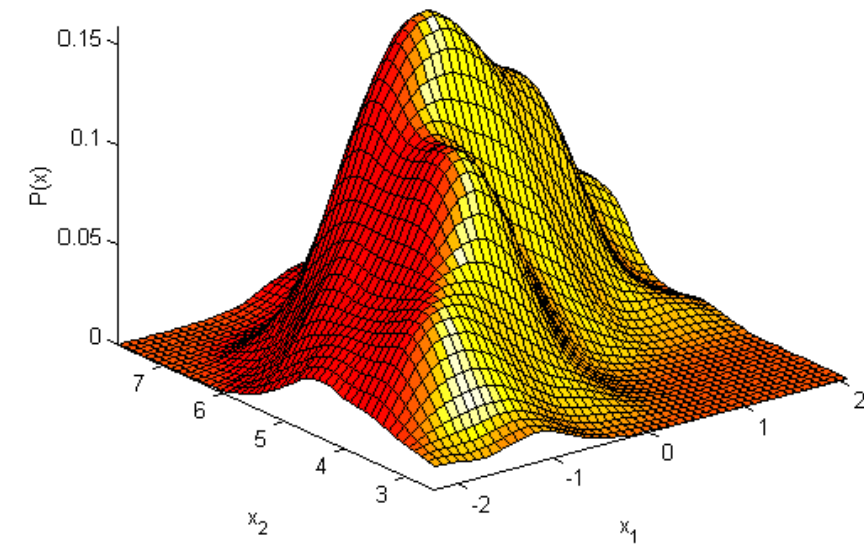
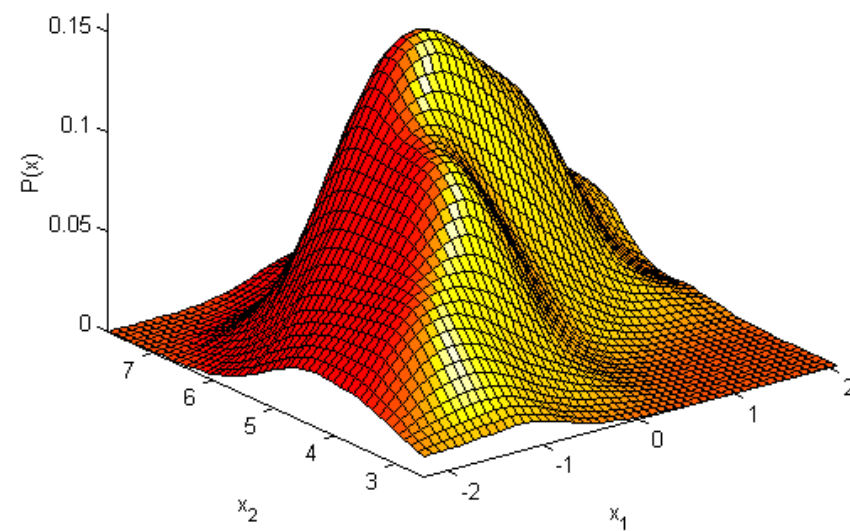
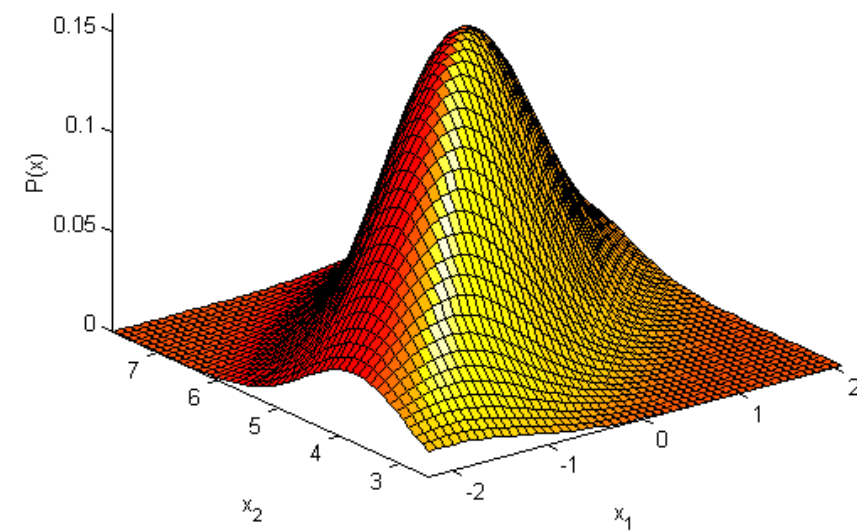
# Effect of the kernel bandwidth $h$

$$p(x) = \frac{1}{m} \sum_i^m \frac{1}{h} K\left(\frac{x^i - x}{h}\right)$$





# Two-dimensional example



# What is the best kernel bandwidth?

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- Silverman's rule of thumb: If using the Gaussian kernel, a good choice for  $h$  is

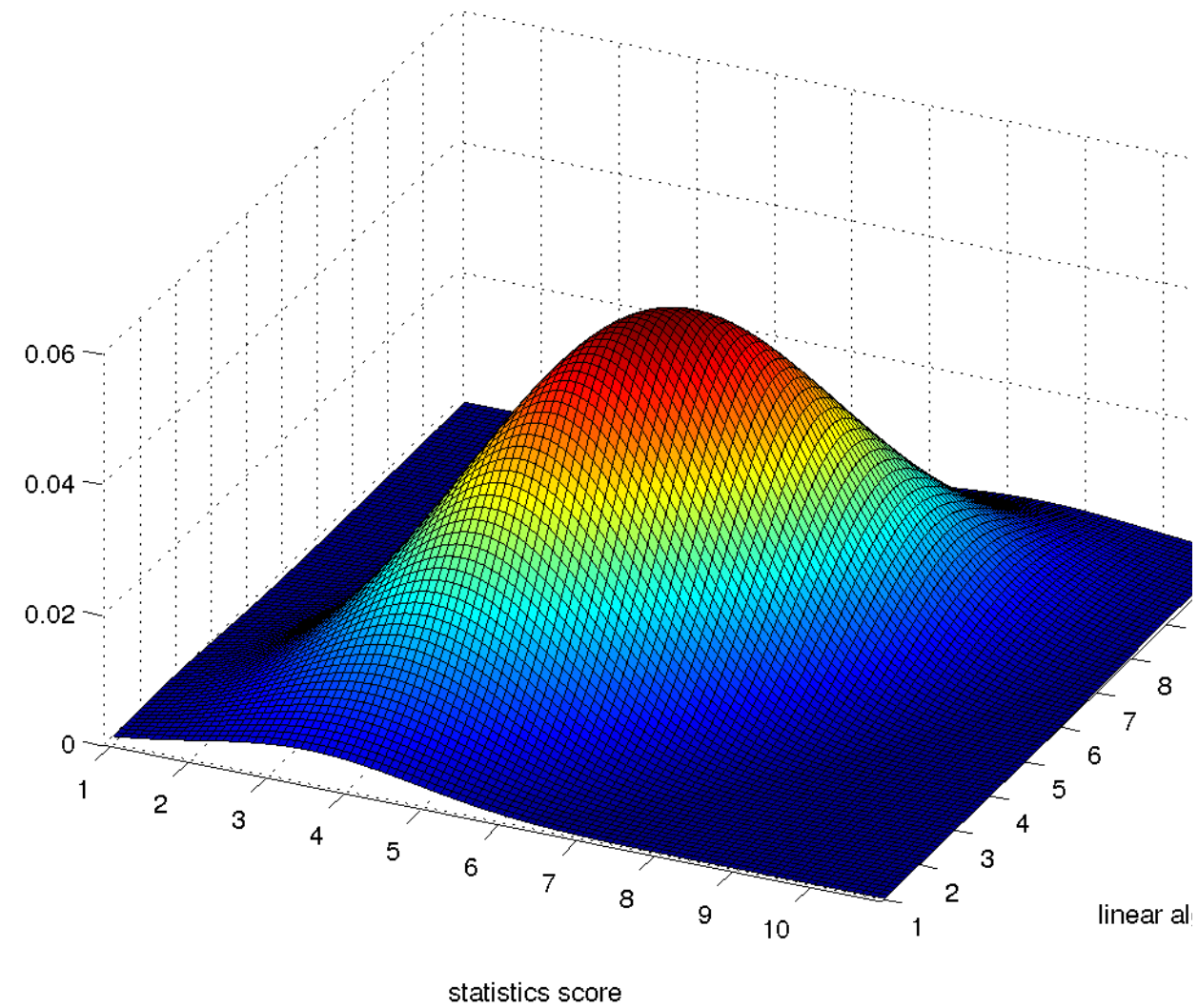
$$h \approx 1.06 \hat{\sigma} m^{-1/5}$$

where  $\hat{\sigma}$  is the standard deviation of the samples

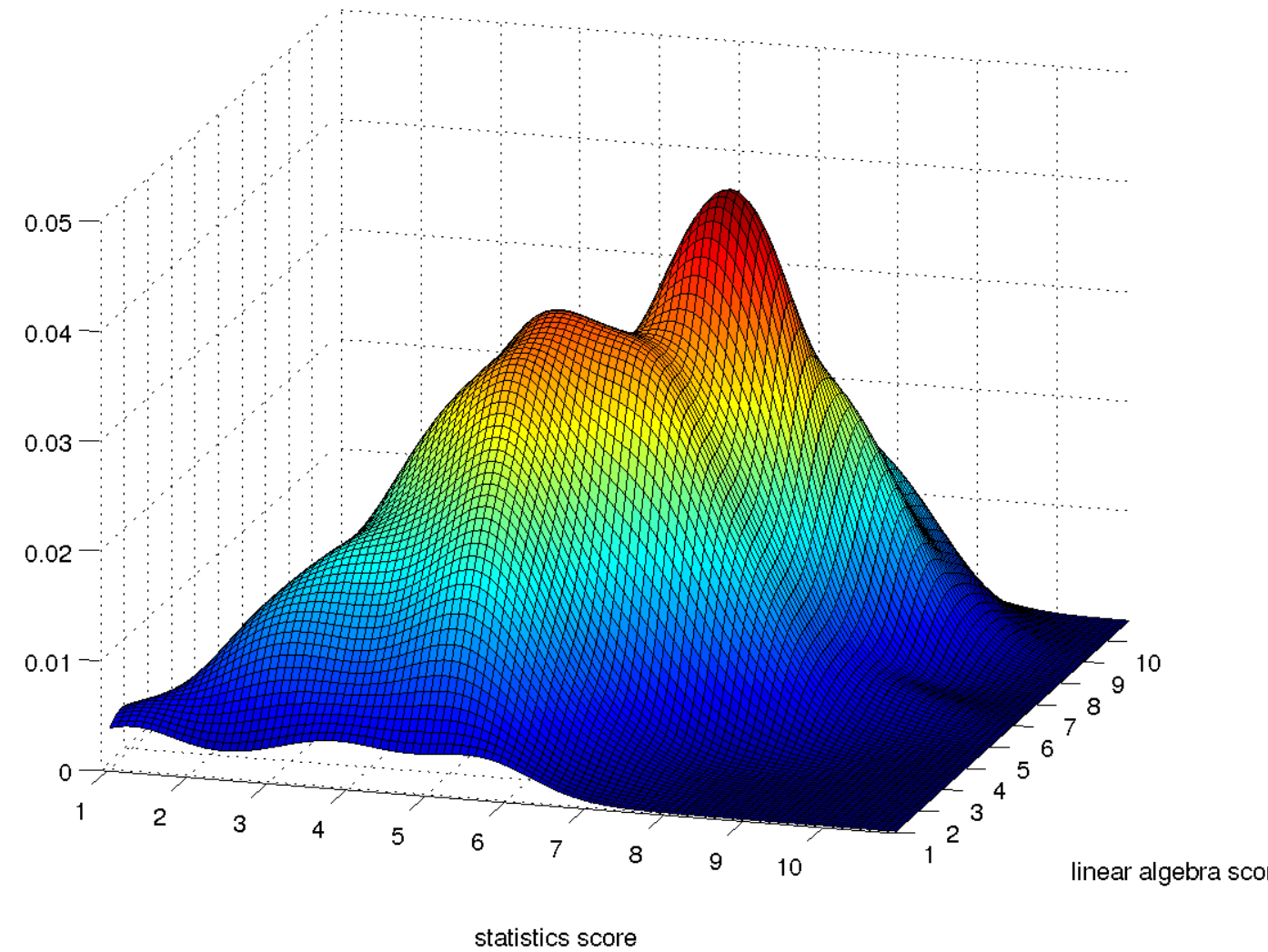
- A better but more computational intensive approach:
  - Randomly split the data into two sets
  - Obtain a kernel density estimate for the first
  - Measure the likelihood of the second set
  - Repeat over many random splits and average

# Parametric vs. nonparametric

Gaussian density



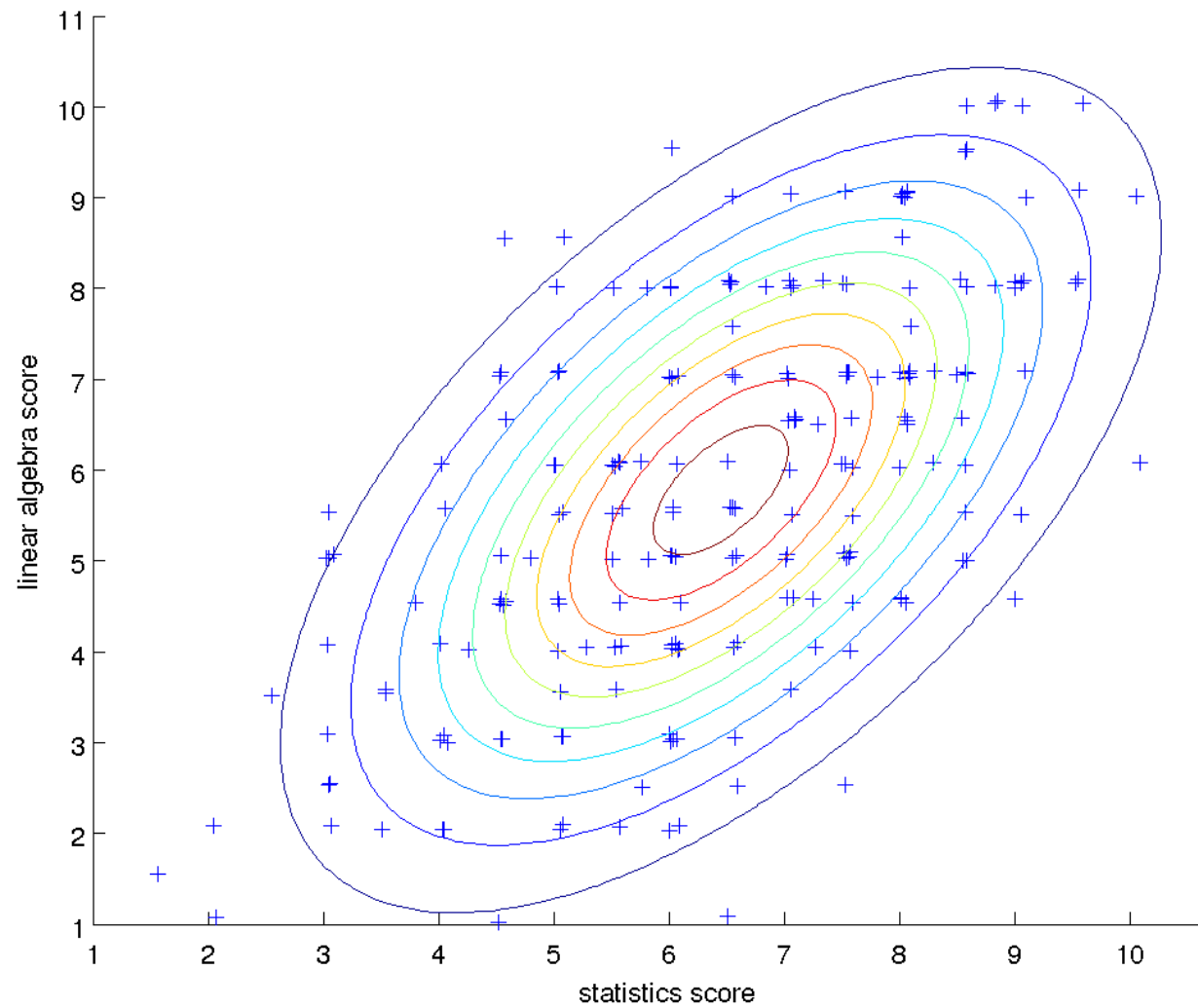
kernel density estimator



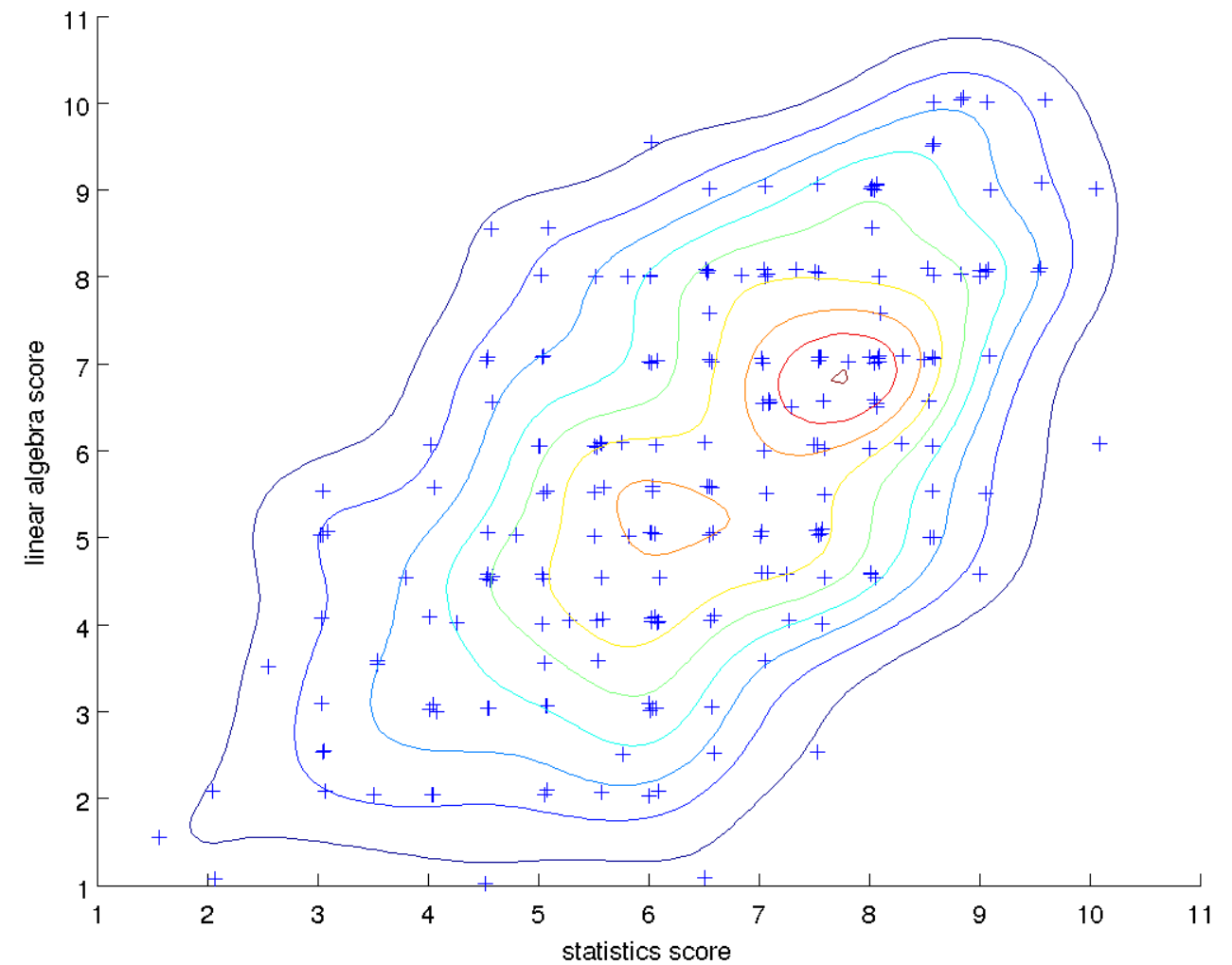


# Parametric vs. nonparametric

statistics vs linear algebra (Gaussian)



statistics vs linear algebra



# Parametric vs. nonparametric

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- Nonparametric models place very mild assumptions on the data distribution and provide good models for complex data

Parametric models rely on very strong (simplistic) distributional assumptions

- Nonparametric models (not histograms) requires storing and computing with the entire data set.

Parametric models, once fitted, are much more efficient in terms of storage and computation.

# Parametric vs. nonparametric

- Data  $x \in R^d$  with **fixed** dimension  $d$
- Given  $m$  training data points  $\{x^1, x^2 \dots, x^m\}$
- Partition  $n$  bin in each dimension

| Aspects               | Gaussian           | Histogram            | KDE                    |
|-----------------------|--------------------|----------------------|------------------------|
| Flexible              | Not                | Yes                  | Yes                    |
| Assumption            | Strong             | Not                  | Not                    |
| Parameter number      | Fixed              | Increase with $n$    | Increase with $m$      |
| Memory requirement    | $d+d^2$            | $n^d$                | $md$                   |
| Training computation  | Closed form        | Binning and Counting | Nothing                |
| Test computation      | Plug in formula    | Find the bin         | Evaluate $m$ functions |
| Statistical guarantee | Only Gaussian case | Arbitrary (worse)    | Arbitrary (better)     |

# Classification using density estimation

$$p(x, y) = p(x|y)p(y) = p(y|x)p(x)$$

- Simple binary classifier for input  $x^i \in R^d$  and label  $y^i \in \{0,1\}$

- Step I: use label to estimate  $p(y = 0)$  and  $p(y = 1)$  *prior*

- Step II: divide your data according to the value of  $y$ , and estimate

$$p(x) = p(x|y=0)p(y=0) + p(x|y=1)p(y=1)$$

$$p(x|y=0) \text{ and } p(x|y=1) = \int p(x, y) dy$$

- Step III: Classify a new test point  $x$  as

$$g(x) := \begin{cases} 1, & \frac{p(x|y=1)p(y=1)}{p(x|y=0)p(y=0)} > 1 \\ 0, & \text{otherwise} \end{cases}$$

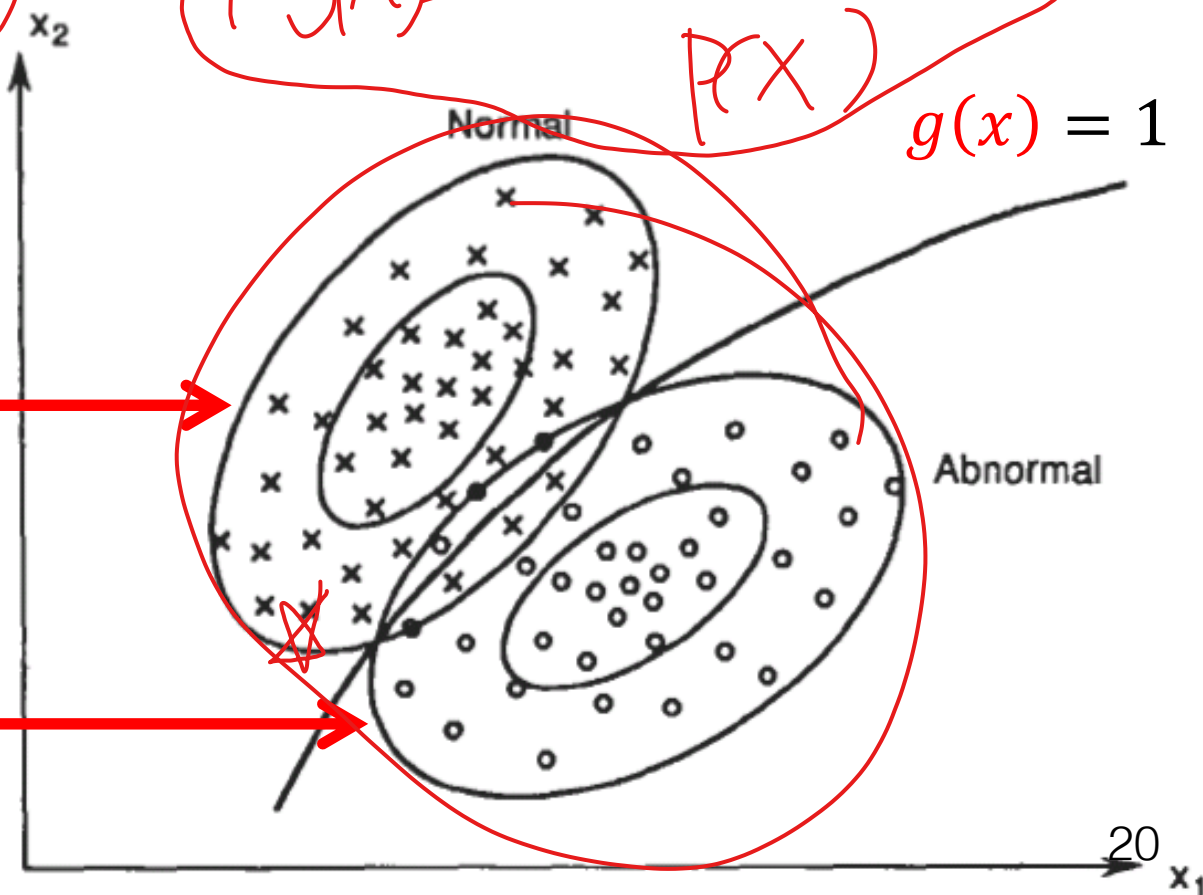
$$p(y|x) = \frac{p(x|y)p(y)}{p(x)}$$

$$g(x) = 1$$

$$g(x) = \frac{p(y=1|x)}{p(y=0|x)}$$

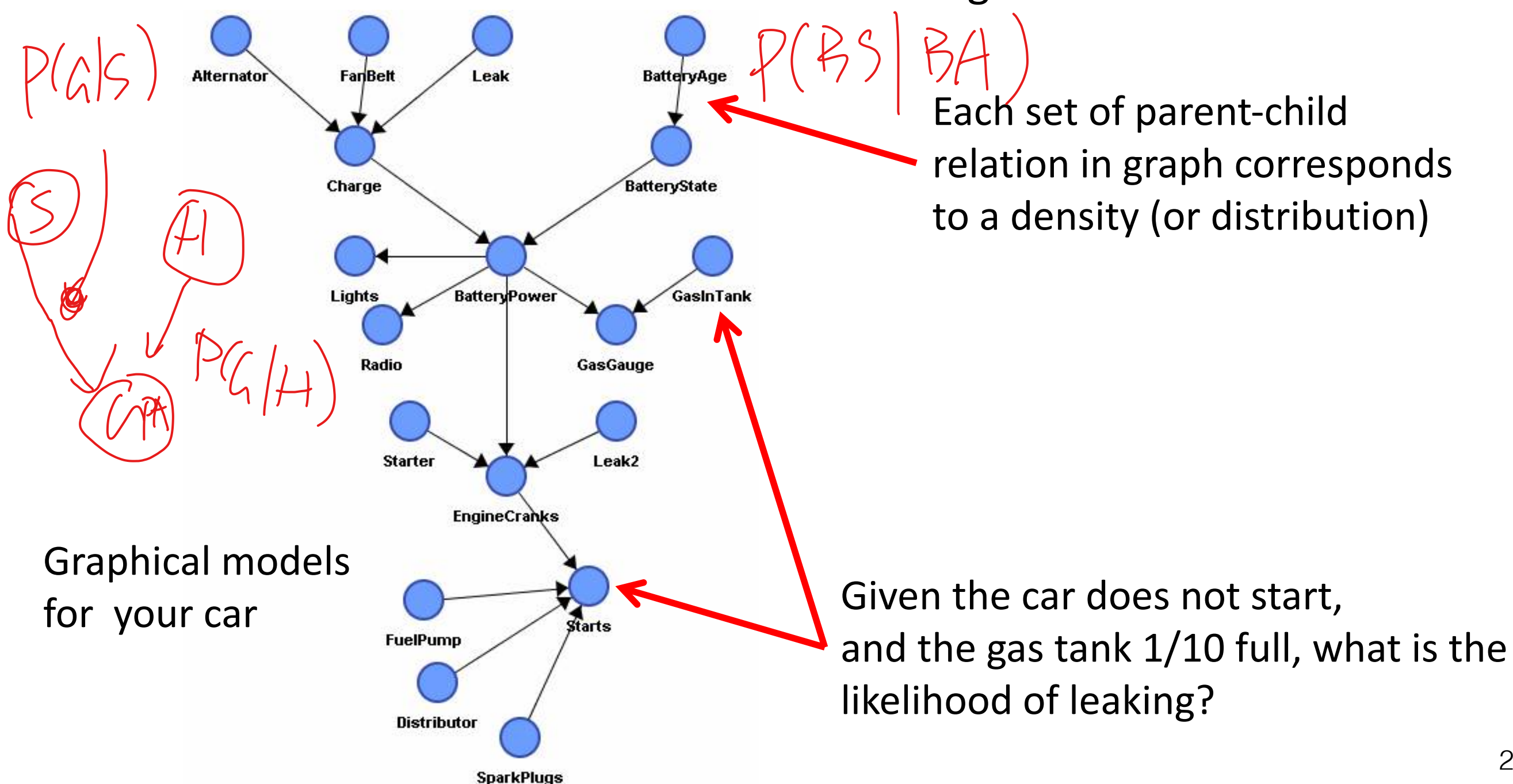
$$p(x|y=0)$$

$$p(x|y=1)$$



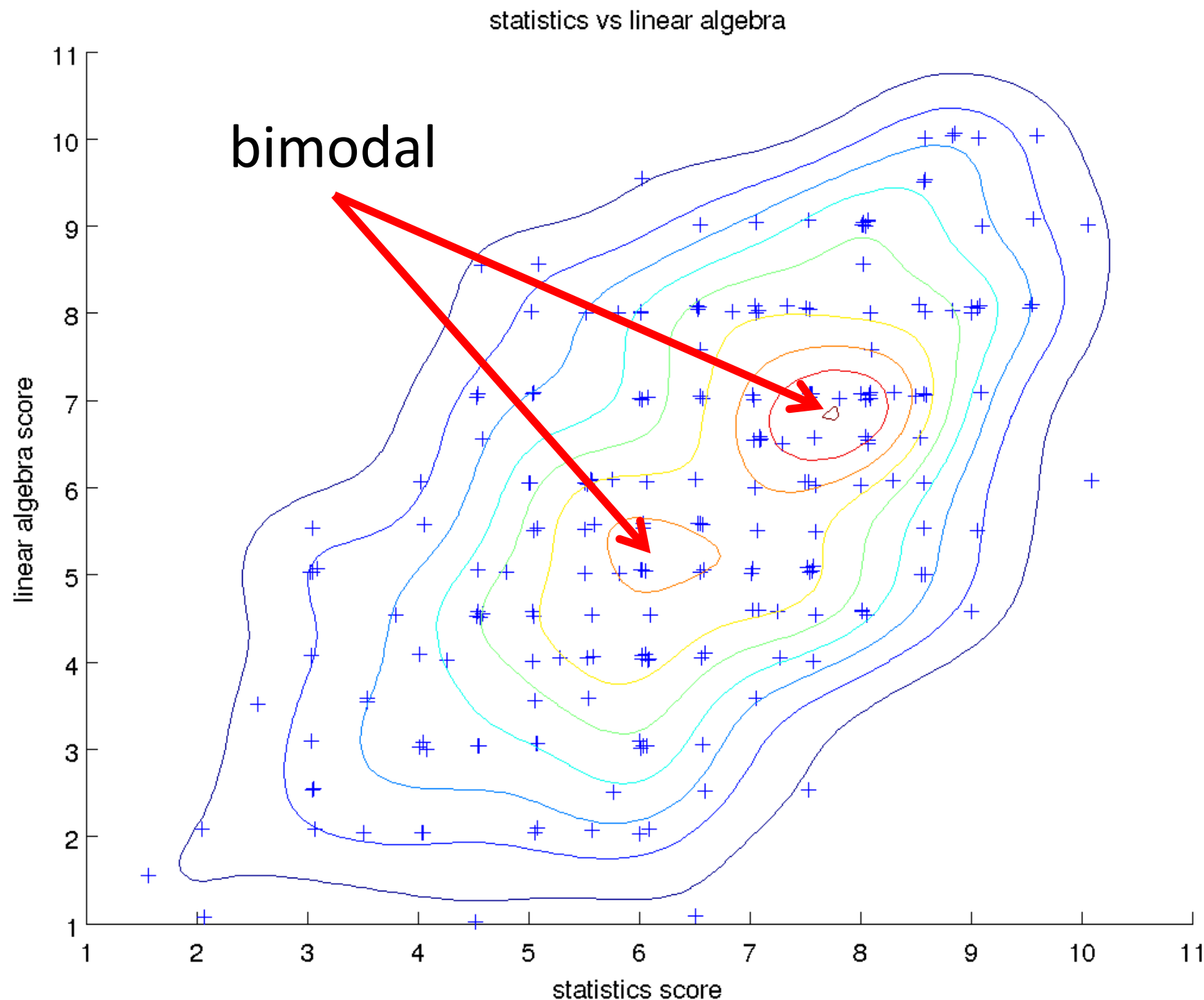
# Graphical models with density estimation

- (Probabilistic) graphical models
  - Use graph to describe the relation of many variables
  - Reason about some unknown variable given known variables



# test scores revisited

- Can we have a more flexible model than Gaussian, but less parameter than KDE?





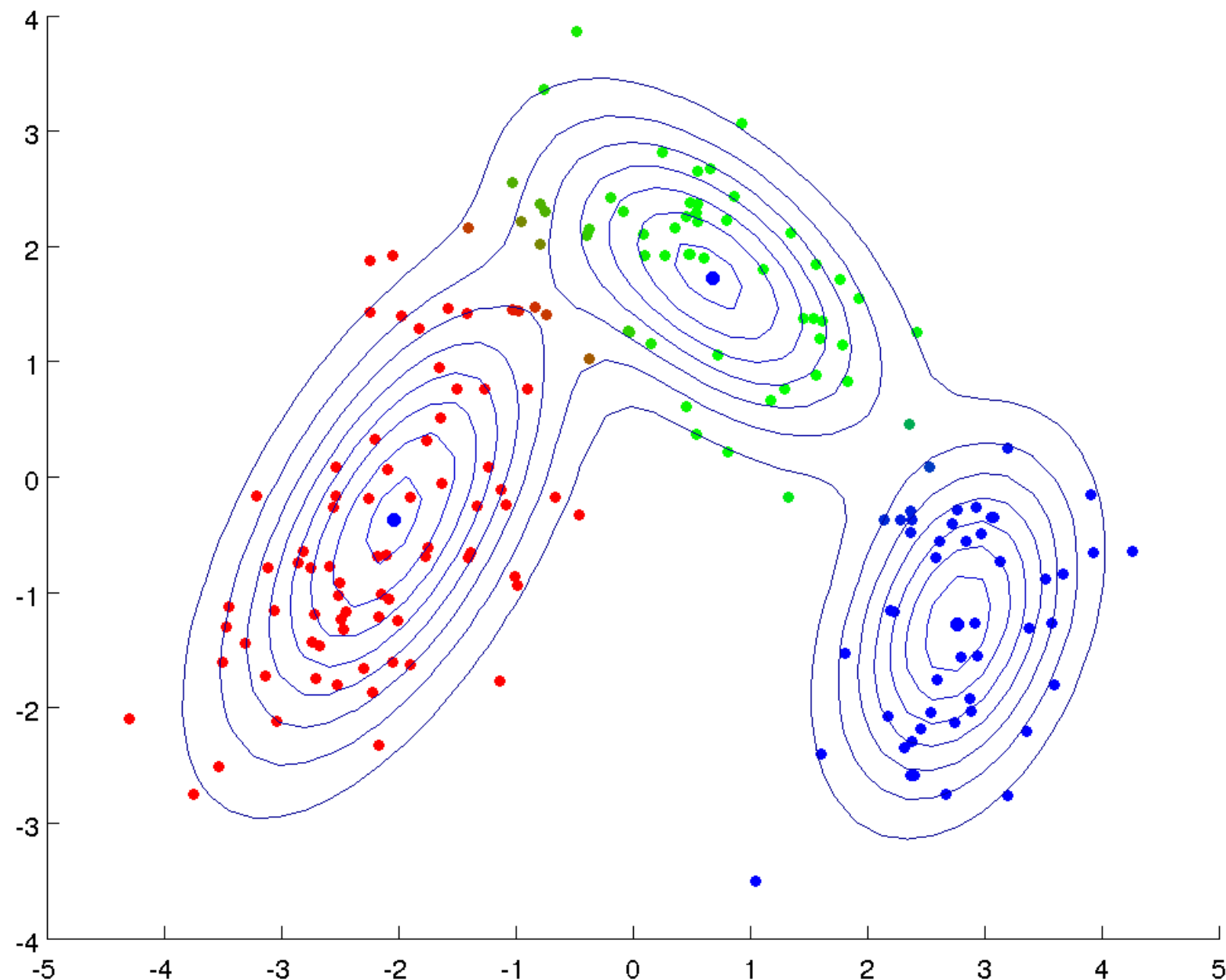
# Wine data set <http://archive.ics.uci.edu/ml/datasets/Wine+Quality>

- These data are the results of a chemical analysis of wines grown in the same region in Italy but derived from three different cultivars. Feature include
  - 1) Alcohol
  - 2) Malic acid
  - 3) Ash
  - 4) Alcalinity of ash
  - 5) Magnesium
  - 6) Total phenols
  - 7) Flavanoids
  - 8) Nonflavanoid phenols
  - 9) Proanthocyanins
  - 10) Color intensity
  - 11) Hue
  - 12) OD280/OD315 of diluted wines
  - 13) Proline



# Demo: test\_wine.m

- First run PCA to reduce the dimension to 2
- Clear cluster structure
- Can we fit 3 Gaussians?



# Gaussian mixture model

$$\log N = - \left( \right)^2$$

$$\log(\pi_1 N_1 + \pi_2 N_2) = \log \sum \exp$$

- A density model  $p(X)$  may be multi-modal: model it as a mixture of uni-modal distributions (e.g. Gaussians)

$$\mathcal{N}(X|\mu_k, \Sigma_k) := \frac{1}{|\Sigma|^{1/2} (2\pi)^{d/2}} \exp\left(-\frac{1}{2} (X - \mu)^T \Sigma^{-1} (X - \mu)\right)$$

$$\int p(x) dx = \sum_K \pi_k \int \mathcal{N}(x) dx$$

- Consider a mixture of  $K$  Gaussians

$$p(X) = \sum_{k=1}^K \pi_k \mathcal{N}(X|\mu_k, \Sigma_k)$$

$\geq 0$

mixing proportion

mixture Component

$$\sum_K \pi_k = 1$$

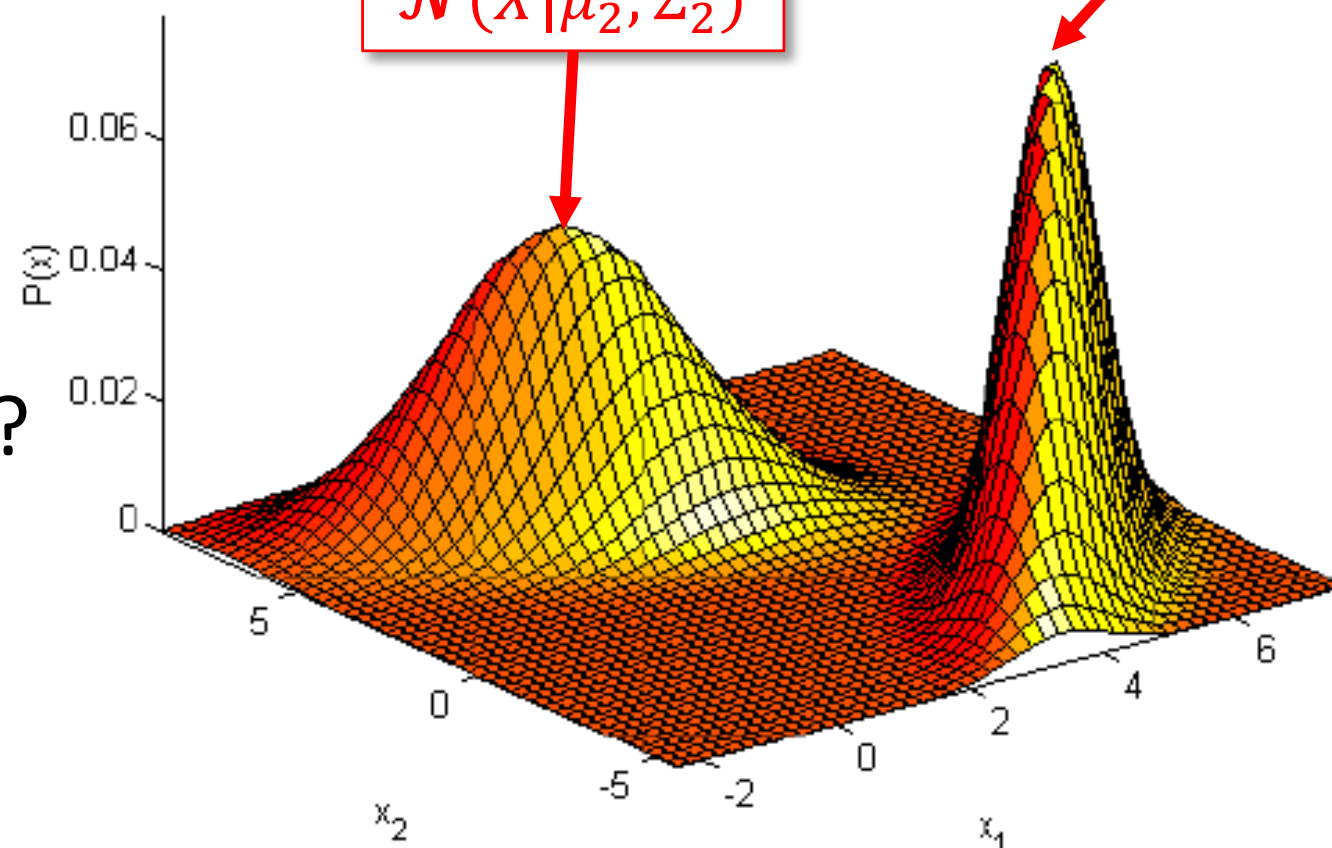
$$\mathcal{N}(X|\mu_1, \Sigma_1)$$

$$\mathcal{N}(X|\mu_2, \Sigma_2)$$

- Parametric or nonparametric?

$$\sum_{k=1}^K \pi_k = 1$$

- Learn  $\pi_k \in (0,1), \mu_k, \Sigma_k$ ;



# Image a generative process for data points

- For each data point  $x^i$ :
  - Randomly choose a mixture component,  $z^i = \{1, 2, \dots, K\}$ , with probability  $\pi_{z^i}$   
 $\pi_{z=1} = 0.50$   $\pi_{z=2} = 0.5$
  - Then sample the actual value of  $x^i$  from a Gaussian distribution  $\mathcal{N}(x | \mu_{z^i}, \Sigma_{z^i})$

- Joint distribution over  $p(x, z)$

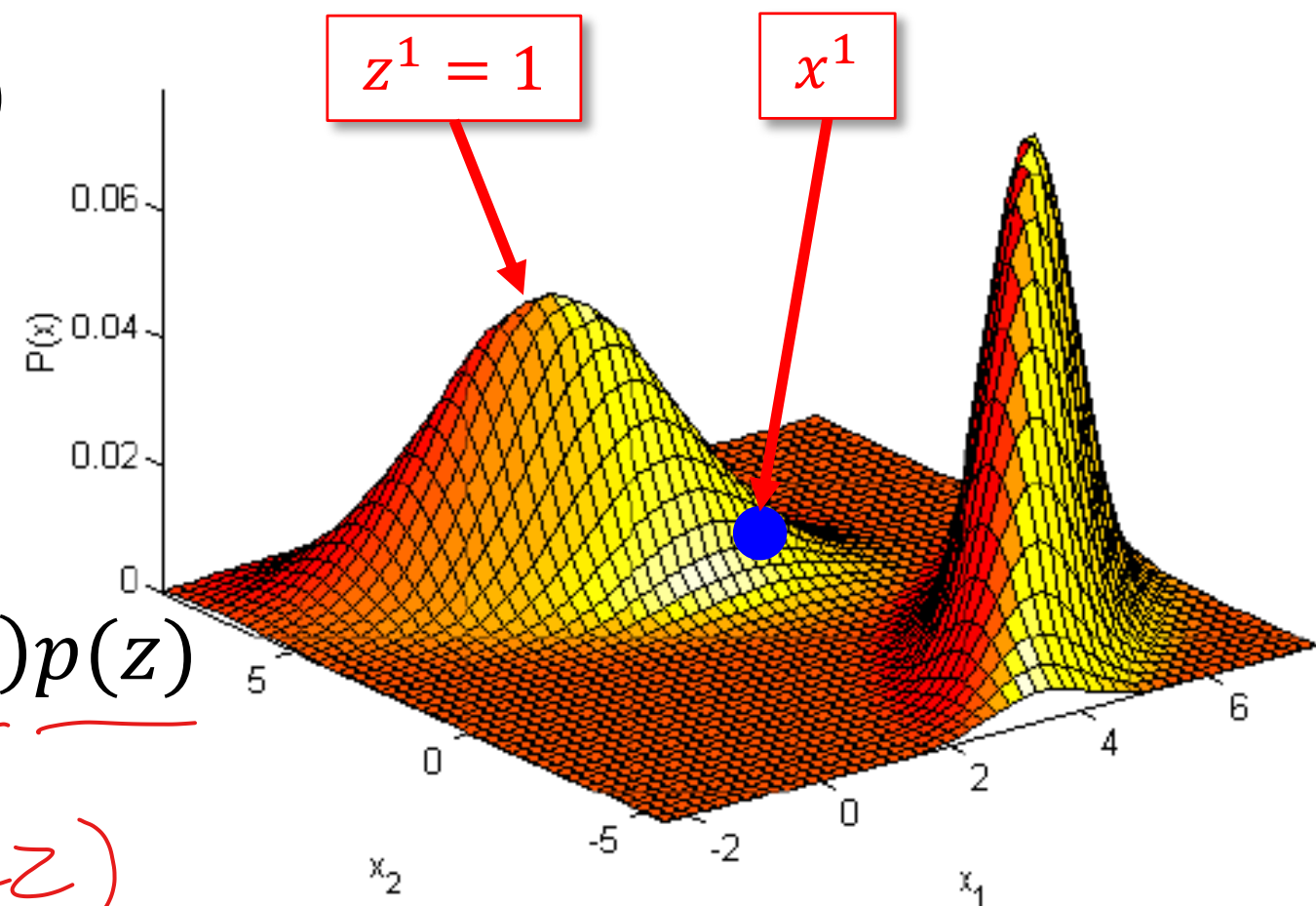
$$p(x, z) = \pi_z \mathcal{N}(x | \mu_z, \Sigma_z)$$

$$= p(x|z)p(z)$$

- Marginal distribution  $p(x)$

$$p(x) = \sum_{z=1}^K p(x, z) = \sum_{z=1}^K p(x|z)p(z)$$

$$= p(x|z=1)p(z=1) + p(x|z=2)p(z=2)$$



- How to learn?
- Maximum likelihood learning (let  $\theta = (\pi_k, \mu_k, \Sigma_k), k = 1 \dots K$ )

$$\theta^* = \operatorname{argmax} l(\theta; D) = \log \prod_{i=1}^m p(x^i)$$

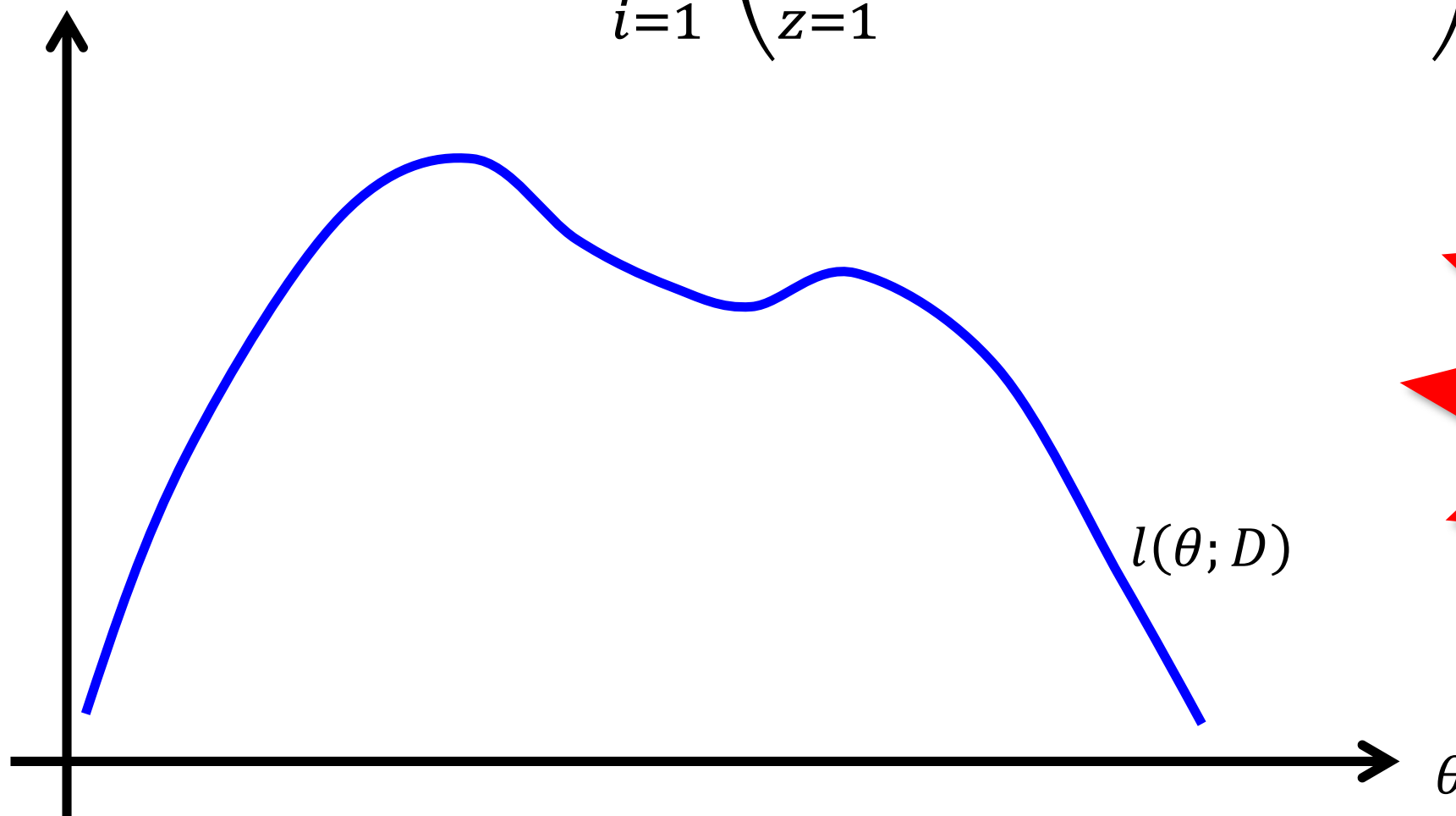
- Use our generative process

$$\begin{aligned} l(\theta; D) &= \log \prod_{i=1}^m \left( \sum_{z^i=1}^K p(x^i, z^i | \theta) \right) \\ &= \log \prod_{i=1}^m \left( \sum_{z^i=1}^K p(x^i | \mu_{z^i}, \Sigma_{z^i}) p(z^i | \pi) \right) \end{aligned}$$

# Why is learning hard?

- With latent variables  $z$ , likelihood of the data becomes

$$l(\theta; D) = \log \prod_{i=1}^m \left( \sum_{z^i=1}^K p(x^i | \mu_{z^i}, \Sigma_{z^i}) p(z^i | \pi) \right)$$
$$= \log \prod_{i=1}^m \left( \sum_{z=1}^K \pi_{z^i} \mathcal{N}(x | \mu_{z^i}, \Sigma_{z^i}) \right)$$



Nonconvex  
Difficult!

## When $z$ is given

- Let us introduce a binary variable  $\tau_k^i$  to indicate whether  $z^i = k$

$$\begin{aligned} l(\theta; D) &= \log \prod_{i=1}^m \left( \sum_{z^i=1}^K p(x^i | \mu_{z^i}, \Sigma_{z^i}) p(z^i | \pi) \right) \\ &= \log \prod_{i=1}^m \left( \sum_{z=1}^K \pi_{z^i} \mathcal{N}(x | \mu_{z^i}, \Sigma_{z^i}) \right) \\ &= \sum_{i=1}^m \sum_{k=1}^K \tau_k^i \log \pi_k \mathcal{N}(x_i | \mu_k, \Sigma_k) \\ &= \sum_{i=1}^m \sum_{k=1}^K \tau_k^i \left[ \log \pi_k - (x^i - \mu_k)^\top \Sigma_k^{-1} (x^i - \mu_k) + \log \Sigma_k + c \right] \end{aligned}$$

## When $z$ is given

- Let us introduce a binary variable  $\tau_k^i$  to indicate whether  $z^i = k$

$$l(\theta; D) = \sum_{i=1}^m \sum_{k=1}^K \tau_k^i \left[ \log \pi_k - (x^i - \mu_k)^\top \Sigma_k (x^i - \mu_k) + \log \Sigma_k + c \right]$$

- It is now easy to show that, the maximum likelihood estimation of the parameters are

$$\begin{aligned} \pi_k &= \frac{\sum_i \tau_k^i}{m}, & \mu_k &= \frac{\sum_i \tau_k^i x^i}{\sum_i \tau_k^i} \\ \Sigma_k &= \frac{\sum_i \tau_k^i (x^i - \mu_k)(x^i - \mu_k)^\top}{\sum_i \tau_k^i} \\ & \quad (k = 1 \dots K) \end{aligned}$$



# When $z$ is not given

- We can guess which Gaussian  $x^i$  comes from by computing the posterior probability

$$p(z^i = k | D, \mu, \Sigma) = \frac{\pi_k \mathcal{N}(x_i | \mu_k, \Sigma_k)}{\sum_{k'=1}^K \pi_{k'} \mathcal{N}(x_i | \mu_{k'}, \Sigma_{k'})} \quad (k = 1 \dots K, i = 1 \dots m)$$

- Bayes rule

likelihood                      Prior

Prior:  $p(z) = \pi_z$   
Likelihood:  $p(x|z) = \mathcal{N}(x|\mu_z, \Sigma_z)$

$$P(z|x) = \frac{P(x|z)P(z)}{P(x)} = \frac{P(x, z)}{\sum_{z'} P(x, z')}$$

posterior                      normalization constant

## When $z$ is not given

- We can guess which Gaussian  $x^i$  comes from by computing the posterior probability

$$p(z^i = k | D, \mu, \Sigma) = \frac{\pi_k \mathcal{N}(x_i | \mu_k, \Sigma_k)}{\sum_{k'=1}^K \pi_{k'} \mathcal{N}(x_i | \mu_{k'}, \Sigma_{k'})} \quad (k = 1 \dots K, i = 1 \dots m)$$

- We are going to pretend  $p(z_k^i = 1 | D, \mu, \Sigma)$  as  $\tau_k^i$  which should be binary – but now is regarded as “soft” assigning  $x^i$  to  $k$ -th component.

- Associate each data and each component with a  $\tau_k^i$
- Initialize  $(\pi_k, \mu_k, \Sigma_k)$ ,  $k = 1 \dots K$
- Iterate the following two steps till convergence:
  - **Expectation step (E-step)**: update  $\tau_k^i$  given current  $(\pi_k, \mu_k, \Sigma_k)$

$$\tau_k^i = p(z_k^i = 1 | D, \mu, \Sigma) = \frac{\pi_k \mathcal{N}(x_i | \mu_k, \Sigma_k)}{\sum_{k'=1}^K \pi_{k'} \mathcal{N}(x_i | \mu_{k'}, \Sigma_{k'})}$$

$(k = 1 \dots K, i = 1 \dots m)$

- **Maximization step (M-step)**: update  $(\pi_k, \mu_k, \Sigma_k)$  given  $\tau_k^i$

$$\pi_k = \frac{\sum_i \tau_k^i}{m}, \quad \mu_k = \frac{\sum_i \tau_k^i x^i}{\sum_i \tau_k^i}$$
$$\Sigma_k = \frac{\sum_i \tau_k^i (x^i - \mu_k)(x^i - \mu_k)^T}{\sum_i \tau_k^i}$$

$(k = 1 \dots K)$

- We intend to learn the parameters that maximizes the log-likelihood of the data

$$l(\theta; D) = \log \prod_{i=1}^m \left( \sum_{z^i=1}^K p(x^i, z^i | \theta) \right)$$



Nonconvex  
Difficult!

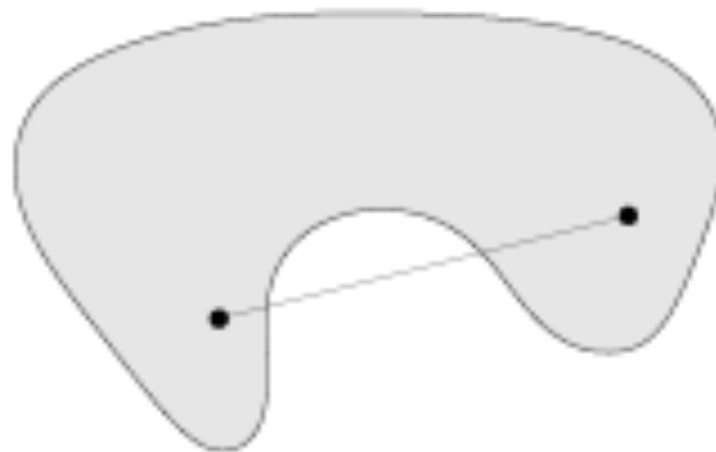
- Expectation step (E-step): What do we take expectation over?

$$l(\theta; D) \geq f(\theta) = E_{q(z^1, z^2, \dots, z^m)} \left[ \log \prod_{i=1}^m p(x^i, z^i | \theta) \right]$$

- Maximization step (M-step): how to maximize?

$$\theta^{t+1} = \operatorname{argmax}_{\theta} f(\theta)$$

- Definition: A set  $A$  is convex, if for every  $0 \leq \alpha \leq 1$  it satisfies
  - $\forall x, y \in A \rightarrow \alpha x + (1 - \alpha)y \in A$
- The line segment between any two points is also in the set.
- Examples of convex and non-convex sets



- Definition: A function  $f: R^n \rightarrow R$  is **convex** if the domain  $\mathbf{dom} f$  is a convex set and if for all  $x, y \in \mathbf{dom} f$ , and  $0 \leq \theta \leq 1$ , we have

$$f(\theta x + (1 - \theta)y) \leq \theta f(x) + (1 - \theta)f(y)$$

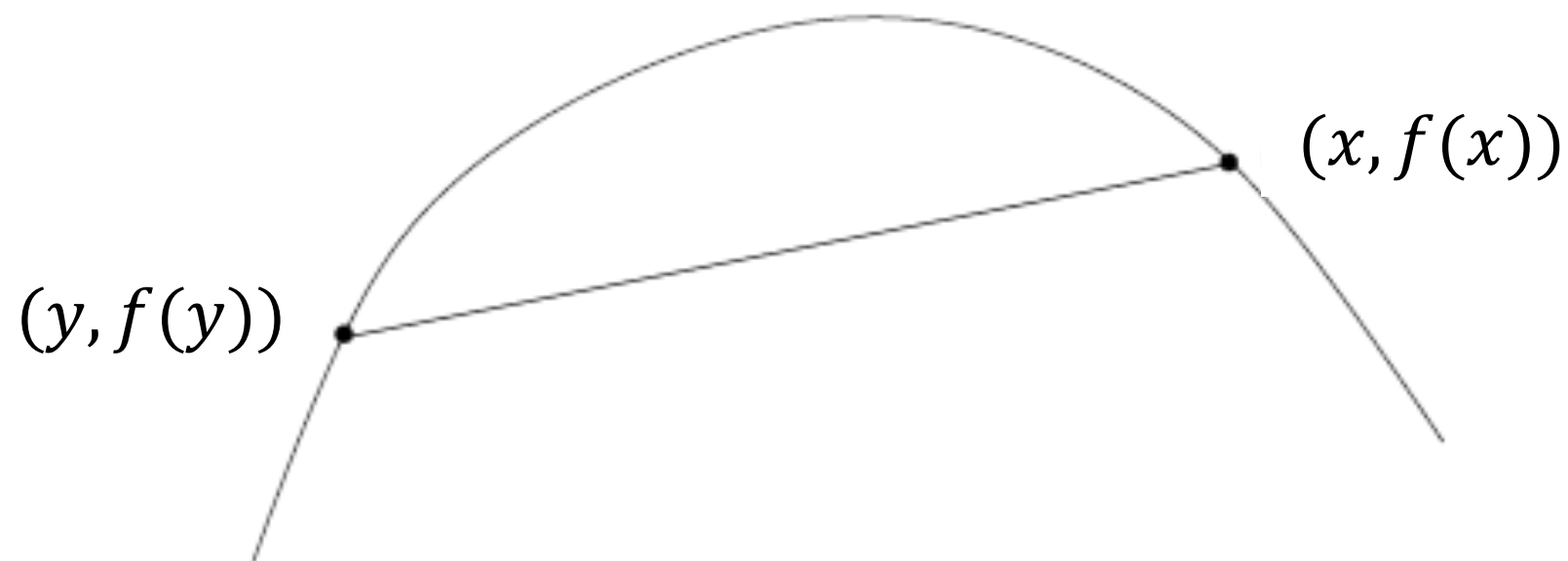
- Geometrically, the line segment between  $(x, f(x))$  and  $(y, f(y))$  lies **above** the graph of  $f$



- Definition: A function  $f: R^n \rightarrow R$  is **concave** if the domain  $\mathbf{dom}f$  is a convex set and if for all  $x, y \in \mathbf{dom}f$ , and  $0 \leq \theta \leq 1$ , we have

$$f(\theta x + (1 - \theta)y) \geq \theta f(x) + (1 - \theta)f(y)$$

- Geometrically, the line segment between  $(x, f(x))$  and  $(y, f(y))$  lies **below** the graph of  $f$





- We intend to learn the parameters that maximizes the log-likelihood of the data

$$l(\theta; D) = \log \prod_{i=1}^m \left( \sum_{z^i=1}^K p(x^i, z^i | \theta) \right)$$



Nonconvex  
Difficult!

- Expectation step (E-step): What do we take expectation over?

$$l(\theta; D) \geq f(\theta) = E_{q(z^1, z^2, \dots, z^m)} \left[ \log \prod_{i=1}^m p(x^i, z^i | \theta) \right]$$

- Maximization step (M-step): how to maximize?

$$\theta^{t+1} = \operatorname{argmax}_{\theta} f(\theta)$$

# Lower bound of log-likelihood

- Log-likelihood  $l(x; \theta) = \log \sum_z p(x, z | \theta)$

$$= \log \sum_z q(z) \frac{p(x, z | \theta)}{q(z)} \text{ (arbitrary } q(z))$$

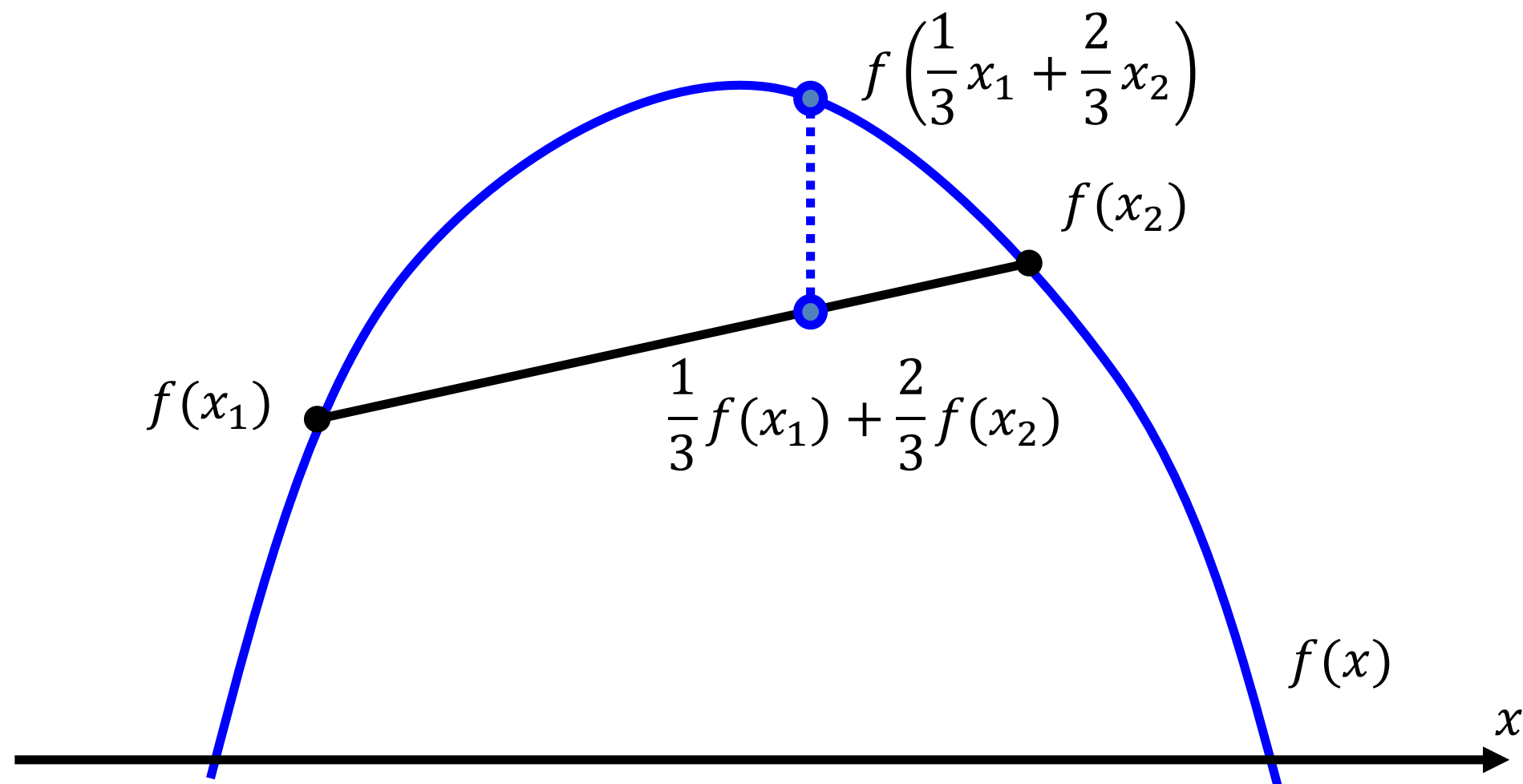
$$\geq \sum_z q(z) \log \frac{p(x, z | \theta)}{q(z)} \text{ (Jensen's inequality } f\left(\sum_i \alpha_i x_i\right) \geq \sum_i \alpha_i f(x_i))$$

$$= \sum_z q(z) \log p(x, z | \theta) - \sum_z q(z) \log q(z)$$

$$= E_{q(z)}[\log p(x, z | \theta)] + H_{q(z)}$$

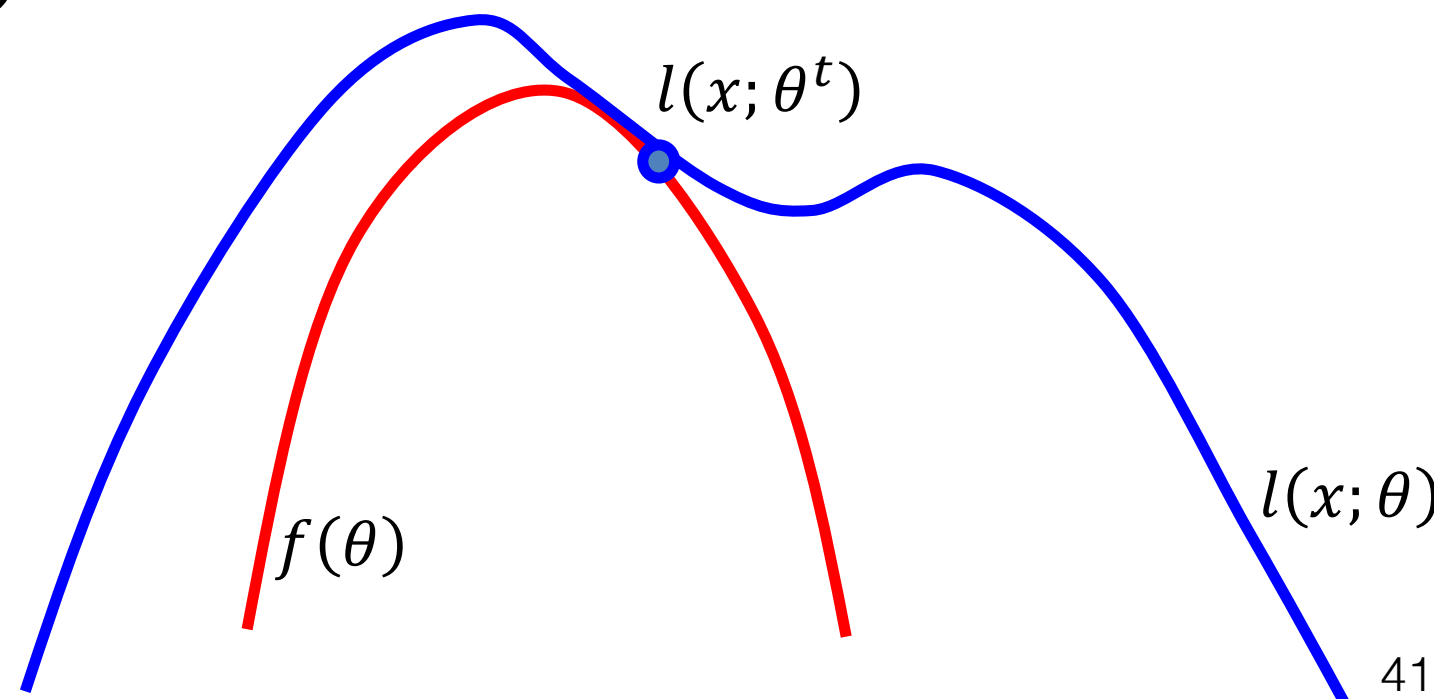
# Jensen's inequality

- For concave function  $f(x)$ 
  - $f(\sum_i \alpha_i x_i) \geq \sum_i \alpha_i f(x_i)$ , where  $\sum_i \alpha_i = 1, \alpha_i \geq 0$
- Most general case: If  $x$  is a random variable, and  $f$  is concave,  
$$f(\mathbf{E}x) \geq \mathbf{E}f(x)$$

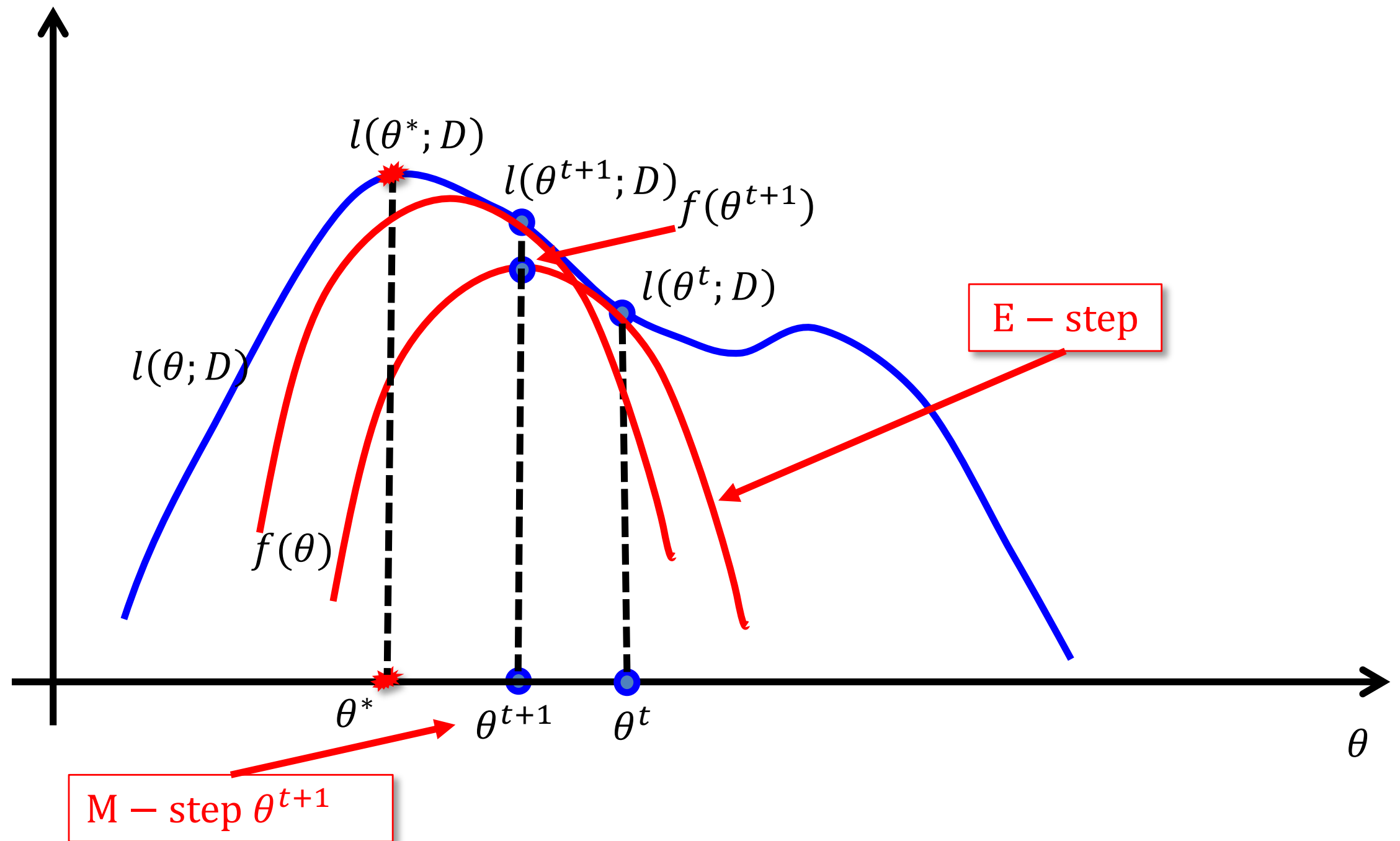


# What attains equality?

- $q(z) = p(z|x, \theta^t)$ : posterior of  $z$  given  $x$  attains the equality at  $\theta^t$
- Let  $F(q, \theta) = \sum_z q(z|x) \log \frac{p(x, z|\theta)}{q(z|x)} \leq l(x; \theta) = \log \sum_z p(x, z|\theta)$
- $F(p(z|x, \theta^t), \theta^t) = \sum_z p(z|x, \theta^t) \log \frac{p(x, z|\theta^t)}{p(z|x, \theta^t)}$
- $= \sum_z p(z|x, \theta^t) \log p(x|\theta^t)$
- $= \log p(x|\theta^t)$
- $= \log \sum_z p(x, z|\theta^t)$



# EM graphically



- Why is  $f(\theta)$  a lower bound? ✓

$$l(\theta; D) \geq f(\theta) = E_{q(z^1, z^2, \dots, z^m)} \left[ \log \prod_{i=1}^m p(x^i, z^i | \theta) \right]$$

- Why  $q(z^1, z^2, \dots, z^m) = \prod_{i=1}^m p(z^i | x^i, \theta^t)$  ✓

- Why will EM converge? ✓

Expectation step (E-step): What do we take expectation over?

$$l(\theta; D) \geq f(\theta) = E_{q(z^1, z^2, \dots, z^m)} \left[ \log \prod_{i=1}^m p(x^i, z^i | \theta) \right]$$

Maximization step (M-step): how to maximize?

$$\theta^{t+1} = \operatorname{argmax}_{\theta} f(\theta)$$

# E-step: what is $q(z^1, z^2, \dots, z^m)$

- $q(z^1, z^2, \dots, z^m)$ : posterior distribution of the latent variables

$$q(z^1, z^2, \dots, z^m) = \prod_{i=1}^m p(z^i | x^i, \theta^t)$$

- For each data point  $x^i$ , compute  $p(z^i = k | x^i)$  for each  $k$

$$\tau_k^i = p(z^i = k | x^i) = \frac{p(z^i = k, x^i)}{\sum_{k'=1..K} p(z^i = k', x^i)}$$

$$= \frac{\pi_k \mathcal{N}(x^i | \mu_k, \Sigma_k)}{\sum_{k'=1..K} \pi_{k'} \mathcal{N}(x^i | \mu_{k'}, \Sigma_{k'})}$$



# E-step: compute the expectation

$$\begin{aligned} f(\theta) &= E_{q(z^1, z^2, \dots, z^m)} \left[ \log \prod_{i=1}^m p(x^i, z^i | \theta) \right] \\ &= \sum_{i=1}^m E_{p(z^i | x^i, \theta^t)} [\log p(x^i, z^i | \theta)] \\ &= \sum_{i=1}^m E_{p(z^i | x^i, \theta^t)} [\log \pi_{z^i} \mathcal{N}(x^i | \mu_{z^i}, \Sigma_{z^i})] \end{aligned}$$

- Expand log of Gaussian  $\log \mathcal{N}(x^i | \mu_{z^i}, \Sigma_{z^i})$

$$\begin{aligned} f(\theta) &= \sum_{i=1}^m E_{p(z^i | x^i, \theta^t)} \left[ \log \pi_{z^i} - (x^i - \mu_{z^i})^\top \Sigma_{z^i}^{-1} (x^i - \mu_{z^i}) + \log \Sigma_{z^i} + c \right] \\ &= \sum_{i=1}^m \sum_{k=1}^K \tau_k^i \left[ \log \pi_k - (x^i - \mu_k)^\top \Sigma_k^{-1} (x^i - \mu_k) + \log \Sigma_k + c \right] \end{aligned}$$



# M-step: maximize $f(\theta)$

- $f(\theta) = \sum_{i=1}^m \sum_{k=1}^K \tau_i^k \left[ \log \pi_k - (x^i - \mu_k)^\top \Sigma_k (x^i - \mu_k) + \log \Sigma_k + c \right]$

- For instance, we want to find  $\pi_k$ , and  $\sum_{i=1}^K \pi_k = 1$

- Form Lagrangian

$$L = \sum_{i=1}^m \sum_{k=1}^K \tau_k^i [\log \pi_k + \text{other terms}] + \lambda (1 - \sum_{i=1}^K \pi_k)$$

- Take partial derivative and set to 0

$$\frac{\partial L}{\partial \pi_k} = \sum_{i=1}^m \frac{\tau_k^i}{\pi_k} - \lambda = 0$$

$$\Rightarrow \pi_k = \frac{1}{\lambda} \sum_{i=1}^m \tau_k^i$$

$$\Rightarrow \lambda = m$$

- Associate each data and each component with a  $\tau_k^i$
- Initialize  $(\pi_k, \mu_k, \Sigma_k)$ ,  $k = 1 \dots K$
- Iterate the following two steps till convergence:
  - **Expectation step (E-step):** update  $\tau_k^i$  given current  $(\pi_k, \mu_k, \Sigma_k)$

$$\tau_k^i = p(z_k^i = 1 | D, \mu, \Sigma) = \frac{\pi_k \mathcal{N}(x_i | \mu_k, \Sigma_k)}{\sum_{k'=1}^K \pi_{k'} \mathcal{N}(x_i | \mu_{k'}, \Sigma_{k'})}$$

$(k = 1 \dots K, i = 1 \dots m)$

- **Maximization step (M-step):** update  $(\pi_k, \mu_k, \Sigma_k)$  given  $\tau_k^i$

$$\pi_k = \frac{\sum_i \tau_k^i}{m}, \quad \mu_k = \frac{\sum_i \tau_k^i x^i}{\sum_i \tau_k^i}$$
$$\Sigma_k = \frac{\sum_i \tau_k^i (x^i - \mu_k)(x^i - \mu_k)^T}{\sum_i \tau_k^i}$$

$(k = 1 \dots K)$

# Expectation-Maximization Iterations

- $k = 1 \text{ or } 2$
- Use  $\tau_1^i$  as the proportion of red, and  $\tau_2^i$  proportion of blue
- Draw only one contour for each Gaussian component

